Lesson 1 — Extra Practice

Slope–Intercept Form and the Graph of a Line

Part A — Is It Linear? Three Answers, Not Two

A relation is linear when it can *be written as $y = mx + b$. Some relations arrive disguised and must be rearranged before you can tell. And a vertical line is a perfectly straight line that still cannot be written as $y = mx + b$ — it needs a row of its own.*

1. Sort every letter below into exactly one of the three rows. Rearrange first where you need to.

(a) $2y = 8x - 14$ (b) $y = x^2 + 1$ (c) $xy = 6$ (d) $x = -3$ (e) $y = \dfrac{3}{x}$ (f) $3x - 5y + 10 = 0$

(g) $y = \sqrt{x} + 1$ (h) $\dfrac{x}{2} + \dfrac{y}{5} = 1$ (i) $y = 2^x$ (j) $y - 3 = -2(x + 1)$ (k) $2x + 7 = 0$ (l) $y = \dfrac{x + 4}{3}$

LINEAR (can be written $y = mx + b$): _____

NON-LINEAR (graph is not a straight line): _____

STRAIGHT LINE, but **not** of the form $y = mx + b$: _____

2. Each pair below looks almost the same but the two members belong in different rows of question 1. Explain the difference in one or two sentences each.

(a) $y = \dfrac{x}{3}$ and $y = \dfrac{3}{x}$ (b) $xy = 6$ and $x + y = 6$ (c) $2y + 7 = 0$ and $2x + 7 = 0$

3. Decide whether each table represents a linear relation. If it does, state m and b and write the equation; if it does not, say which columns prove it. **Warning:** equal first differences in y only settle the matter when the x-values step up by equal amounts.

(a)

x	0	1	2	3
y	-5	-1	3	7

(b)

x	-2	0	2	4
y	9	6	3	0

(c)

x	1	2	3	4
y	2	5	10	17

(d)

x	0	3	6	9
y	4	4	4	4

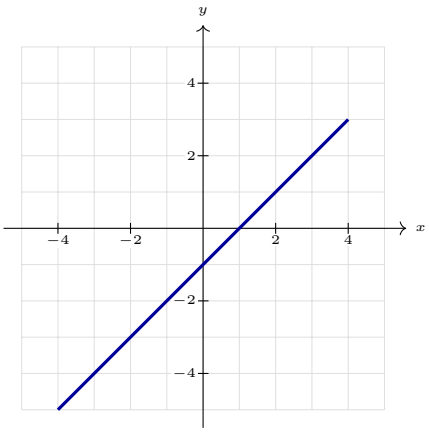
(e)

x	0	1	3	7
y	2	5	11	23

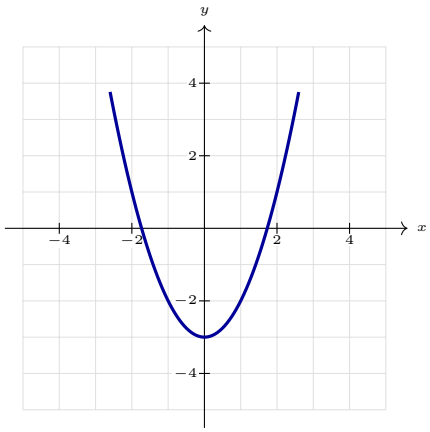
(f) The first differences in the y-row of table (e) are 3, 6 and 12 — nothing like constant — and yet the relation is linear. Explain, in one sentence, what has to be divided by what before "constant first difference" becomes a fair test.

4. State whether each graph shows a linear relation. Justify each answer from the shape of the graph, and for the linear one state m and b.

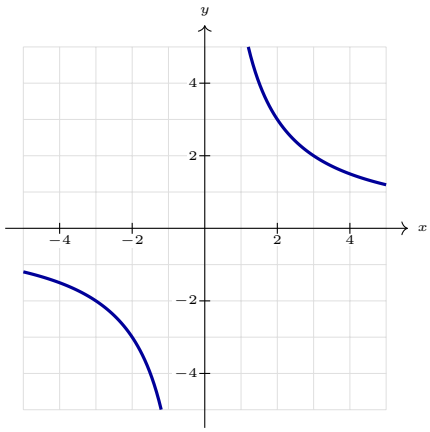
(a)



(b)



(c)



Part B — Reading m and b When the Equation Is Not Solved for y

5. Rearrange each equation into $y = mx + b$, then state the slope and the y-intercept. Leave fractions exact.

(a) $6y = 9x + 24$

(b) $2x + 5y = 15$

(c) $3x - 4y - 8 = 0$

(d) $\dfrac{y}{3} = x - 5$

(e) $y - 2 = \dfrac{2}{3}(x + 9)$

(f) $\dfrac{x}{4} - \dfrac{y}{6} = 1$

(g) $5(y + 1) = 2(x - 3)$

(h) $-3y = 6x - 9$

6. Asked for the slope and y-intercept of $y = 4 - 3x$, a student writes

$$m = 4, \qquad b = -3.$$

(a) Both numbers in the equation appear in the student's answer. Say exactly what they matched to what, and why that matching is wrong.

(b) Rewrite $y = 4 - 3x$ so that it reads in the order $mx + b$, then state the correct m and b.

(c) The student's line and the correct line both pass through one common point. Find it, and explain why checking a second point would have exposed the error.

7. Every equation in general form, $Ax + By + C = 0$, can be converted to slope y-intercept form — almost.

(a) Solve $Ax + By + C = 0$ for y and write m and b in terms of A, B and C.

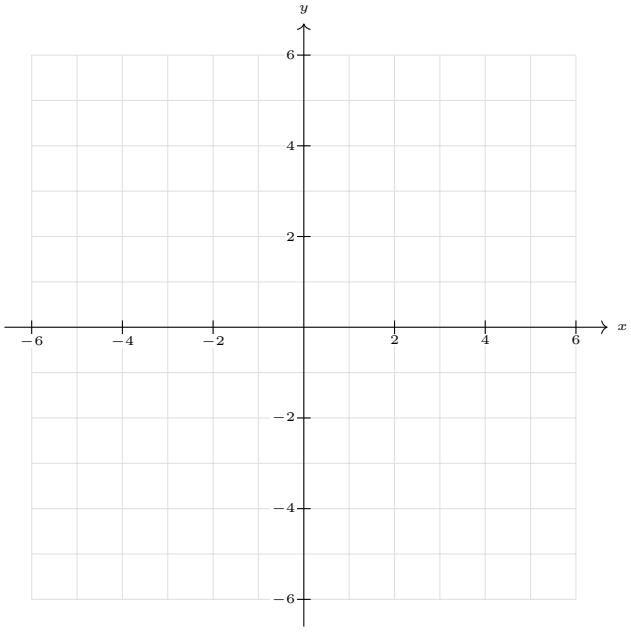
(b) Use your formulas on $5x - 2y + 8 = 0$, then check by rearranging that equation directly.

(c) What goes wrong when $B = 0$? Describe the graph of $3x + 12 = 0$ and say why it has no m and no b.

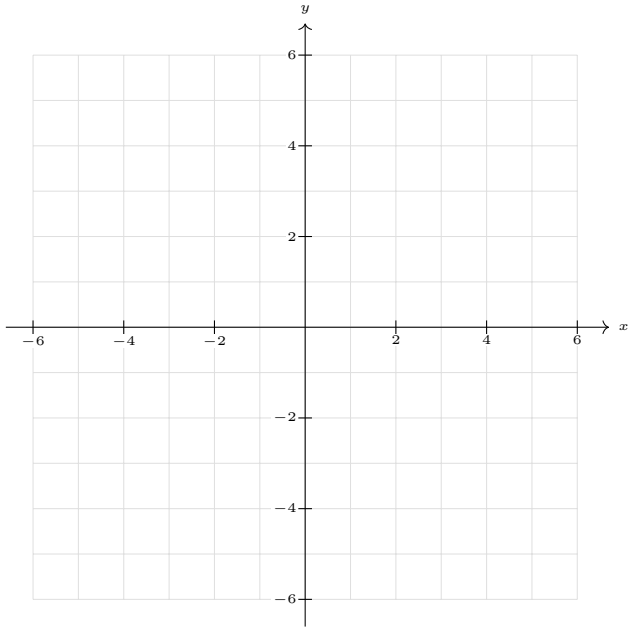
Part C — Graphing From $y = mx + b$

8. For each equation, state the slope and y-intercept, then graph the line on the grid provided. One of the four cannot be put in the form $y = mx + b$ — graph it anyway and say why it resists.

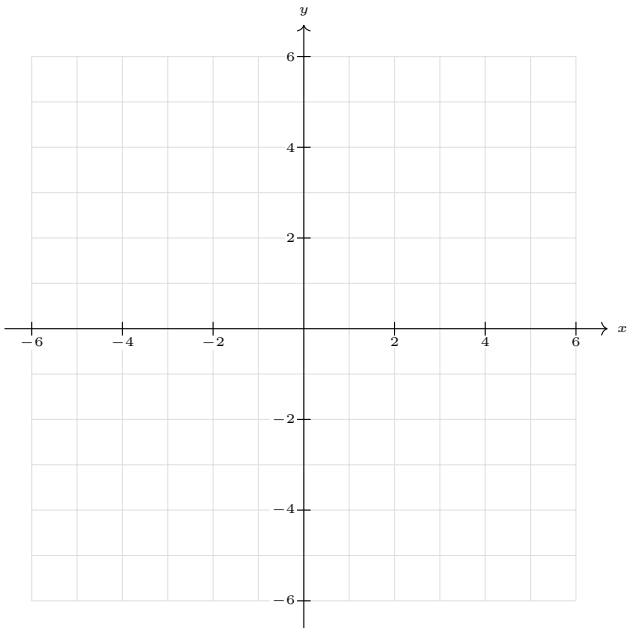
(a) $y = -\frac{2}{3}x + 4$ $m =$ _____ $b =$ _____



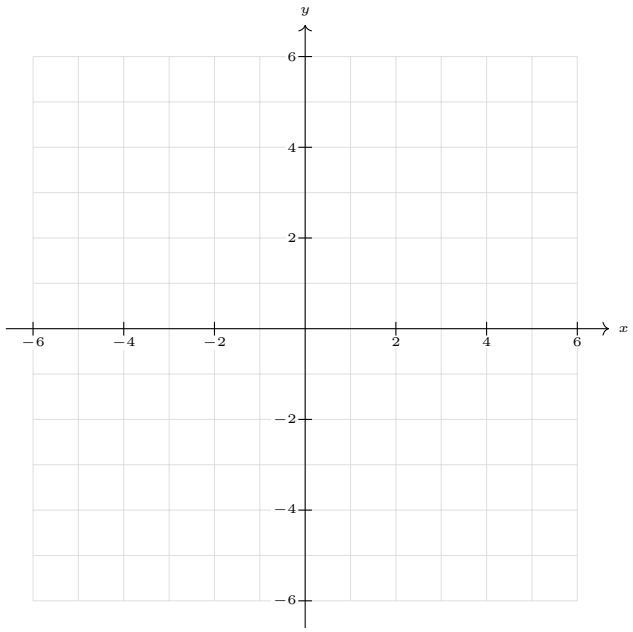
(b) $y = \frac{5}{2}x - 4$ $m =$ _____ $b =$ _____



(c) $2y + 6 = 0$ $m =$ _____ $b =$ _____



(d) $3x - 12 = 0$ $m =$ _____ $b =$ _____



(e) Parts (c) and (d) look alike on the page: one variable, one constant, an equals sign. Explain why one of them has a slope of 0 while the other has no slope value at all.

9. To graph a line of slope $-\frac{2}{3}$ through $(0, 1)$, a student starts at $(0, 1)$ and counts *down* 3 *and right* 2, landing on $(2, -2)$, then joins the two points.

(a) Which part of the fraction is the rise and which is the run? State the correct count.

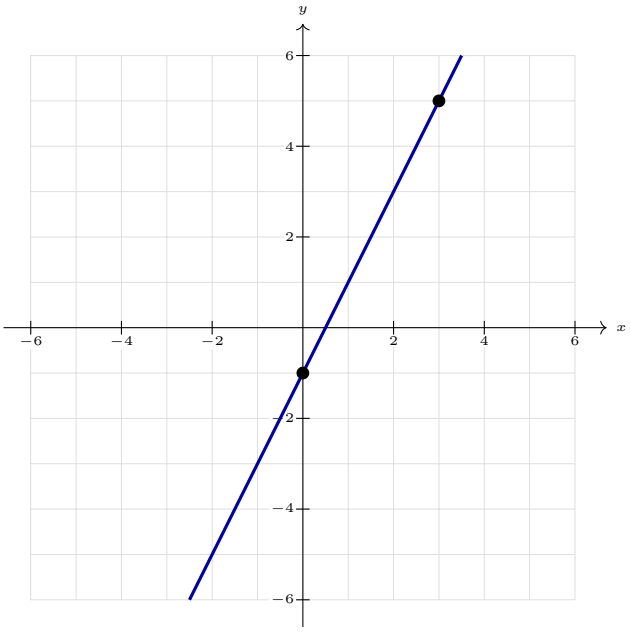
(b) Determine the slope and the equation of the line the student actually drew.

(c) Both lines fall from left to right and both cross the y-axis at $(0, 1)$, so a quick glance will not catch this. Name one point that is on the correct line but not on the student's.

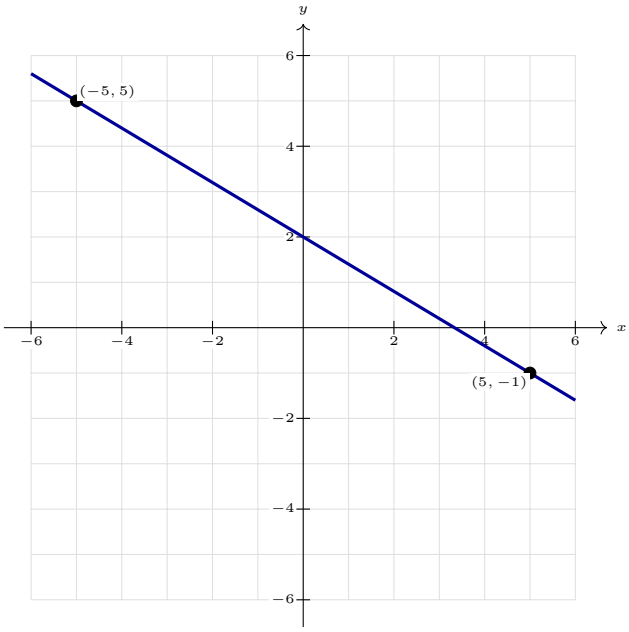
Part D — Backwards: From the Graph to the Equation

10. Each line below is drawn on a grid. Read its slope and y-intercept from the picture and write its equation. In (b) the line does not pass through a lattice point at every step — use the two marked points instead of counting one square at a time.

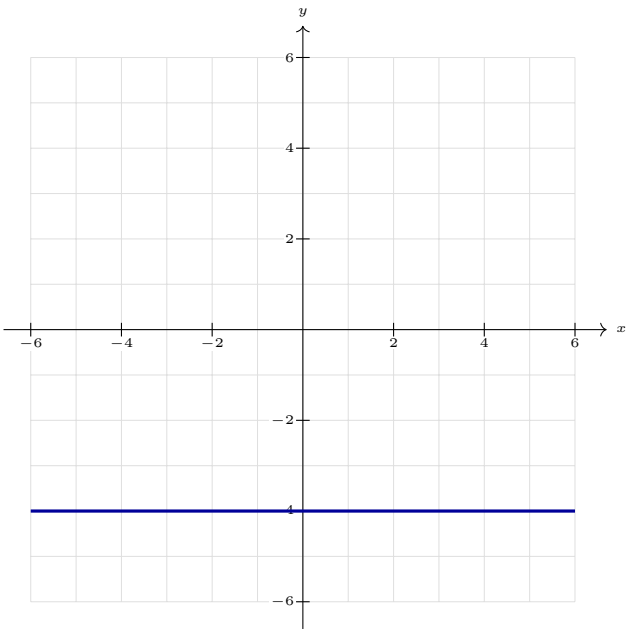
(a)



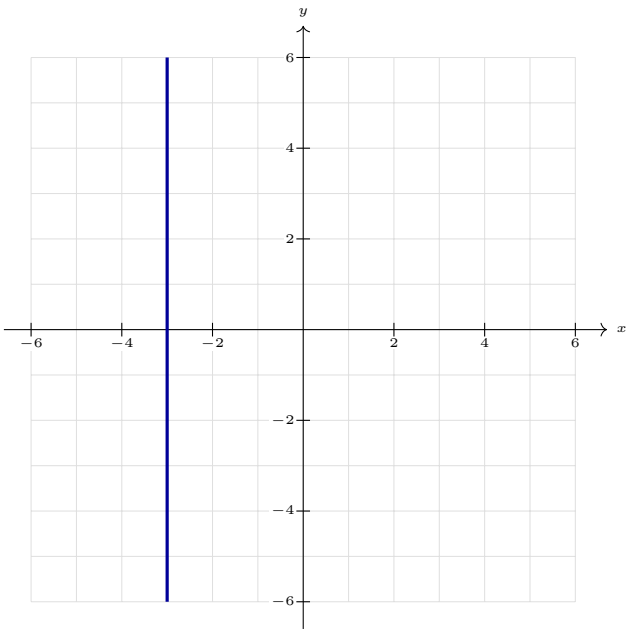
(b)



(c)



(d)

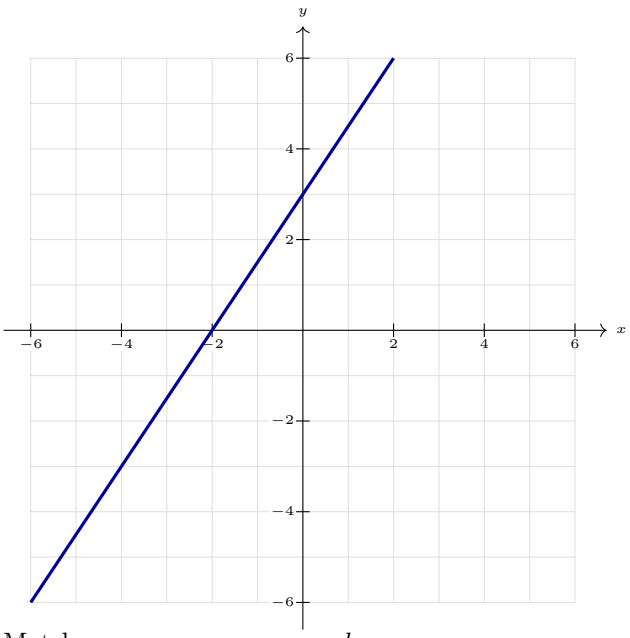


(e) Only one of the four graphs above has no y-intercept to read. Say which, and state what its equation is instead.

11. The four equations below differ only in the *sign* of m or the *value* of b. Match each to one of the graphs **I**–**IV**, and beside each match record the m and b you read off the picture.

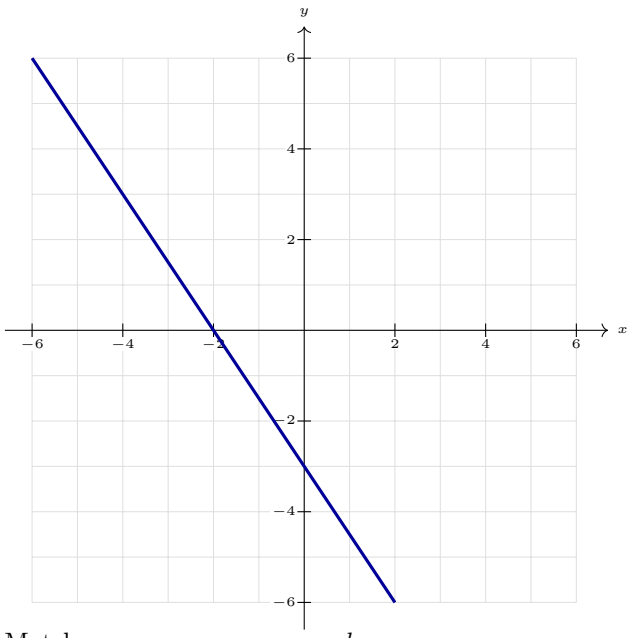
A. $y = \frac{3}{2}x - 3$ **B.** $y = -\frac{3}{2}x - 3$ **C.** $y = \frac{3}{2}x + 3$ **D.** $y = -\frac{3}{2}x + 3$

I.



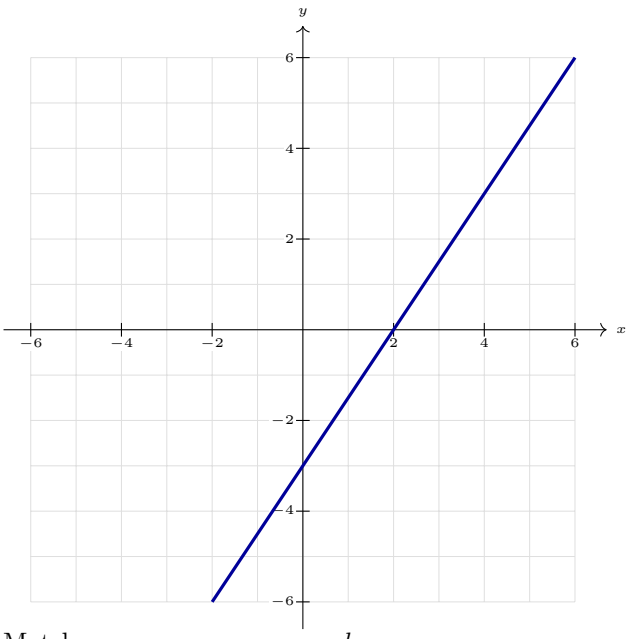
Matches _____; $m =$ _____, $b =$ _____

II.



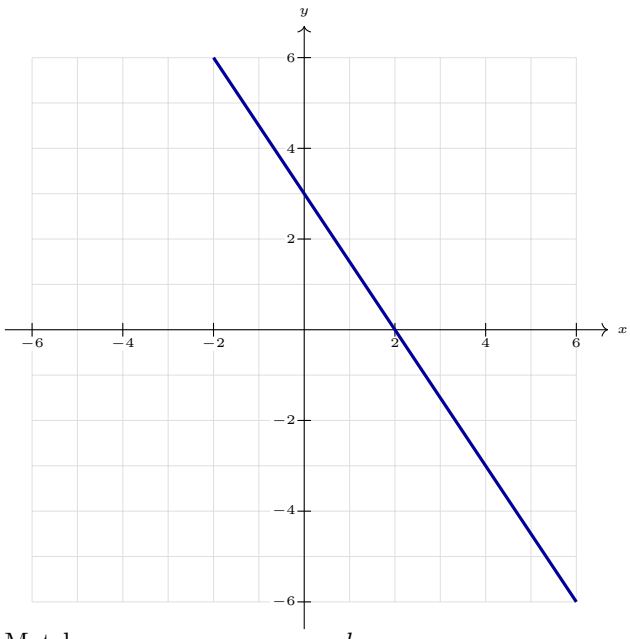
Matches _____; $m =$ _____, $b =$ _____

III.



Matches _____; $m =$ _____, $b =$ _____

IV.



Matches _____; $m =$ _____, $b =$ _____

(e) Two students both answer "$m = \frac{3}{2}$" for graph **II**. Describe the one feature of the picture that settles the sign of m before any counting is done.

Part E — Recovering the Equation From a Table in Context

12. A cell-phone plan charges a fixed monthly fee plus a rate per minute of talk time. The table shows the monthly cost C, in dollars, for n minutes used.

n (min)	0	50	100	150
C (\$)	25.00	31.00	37.00	43.00

(a) Show that the relation is linear, and determine m as an exact rate.

(b) Write the equation in the form $C = mn + b$.

(c) Interpret m in a sentence **with units**, and say what b is in this plan.

(d) Why is reading b off this table easier than reading it off most tables?

(e) Determine the cost of a month with 325 minutes of talk time.

(f) A bill comes to \$55.00. Determine the number of minutes used.

Part F — Stretch and Review

13. The points $(2, 7)$ and $(-1, -2)$ both lie on the line $y = mx + b$. **Do not use the slope formula.**

(a) Substitute each point into $y = mx + b$ to write two equations in the unknowns m and b.

(b) Solve the pair for m and b, and state the equation of the line.

(c) Now check m with the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, and explain where that formula is hiding inside your work in (b).

14. Determine the missing constant in each case.

(a) $y = kx - 3$ passes through $(6, 9)$. Find k.

(b) $y = -4x + c$ passes through $(-2, 11)$. Find c.

(c) $y = \dfrac{2}{5}x + b$ passes through $(10, 1)$. Find b.

15. *(Multiple Choice)* Which line falls from left to right *and* crosses the y-axis below the origin?

A. $y = 3x - 6$ B. $y = -3x + 6$
C. $y = -3x - 6$ D. $y = 3x + 6$

16. *(Multiple Choice)* Which relation is *not* linear?

A. $2x - 3y = 12$ B. $y = \dfrac{x - 5}{2}$
C. $xy = 10$ D. $y = 7$

17. *(Numerical Response)* The line $y = mx + b$ passes through $(-3, -7)$ and $(1, 5)$. The value of y on this line when $x = 12$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 2 — Extra Practice

From Slope to Equation: Building Linear Equations with $y = mx + b$

Part A — When the Point Is Not the y-Intercept

In Lesson 2 every point handed to you sat on the y-axis, so b could simply be read off. Here the point is somewhere else. The method does not change — substitute the slope and the point's coordinates into $y = mx + b$ — but now b is the unknown you solve for. Leave every answer in the form $y = mx + b$ with exact fractions, never decimals.

1. Determine the equation of each line in the form $y = mx + b$.

(a) slope 3, through $(2, -5)$

(b) slope -2, through $(-4, 7)$

(c) slope $\frac{3}{4}$, through $(8, -1)$

(d) slope $-\frac{2}{5}$, through $(-10, 3)$

(e) slope $\frac{5}{6}$, through $(-3, 4)$

(f) slope $-\frac{1}{2}$, through $(7, 0)$

2. Determine the equation of the line through each pair of points, in the form $y = mx + b$ where possible. One of these lines will surprise you — say which, and why.

(a) $(1, 5)$ and $(4, 14)$

(b) $(-2, 9)$ and $(3, -6)$

(c) $(-6, -1)$ and $(2, 3)$

(d) $(5, -2)$ and $(-3, 4)$

(e) $(-4, -7)$ and $(6, -2)$

(f) $(3, -4)$ and $(9, -4)$

3. **Running the equation backwards.** A line has equation $y = \frac{2}{3}x - 5$. Each part below gives you *one* coordinate of a point on this line. Determine the other, and state the full ordered pair.

(a) the point with $x = 12$

(b) the point with $y = -9$

(c) the point with $x = -3$

(d) the point with $y = 1$

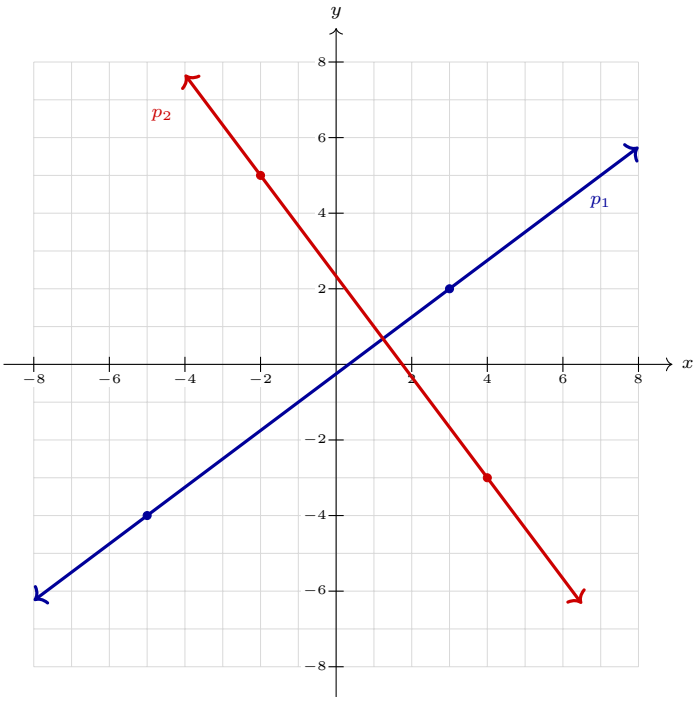
(e) Does $(7, 0)$ lie on this line? Show the substitution that decides it.

(f) the point on the line whose y-coordinate is 0. What is this point called?

(g) Parts (a) and (b) are about the same line, but one of them needed an equation solved and the other did not. Explain which direction is the quick one, and why $y = mx + b$ favours it.

Part B — Building the Equation from a Graph and from a Table

4. Each line below passes through the two marked lattice points. Determine the equation of each line in the form $y = mx + b$.



(a) p_1 passes through $(-5, -4)$ and $(3, 2)$.

(b) p_2 passes through $(-2, 5)$ and $(4, -3)$.

(c) Neither line crosses the y-axis at a grid point. Explain why reading b off this graph could not have given an exact answer, and say what you did instead.

5. Each table below shows a relation. Where the relation is linear, determine its equation in the form $y = mx + b$. Where it is not, say so and explain how the table told you.

(a)

x	-4	-1	2	5	8
y	13	7	1	-5	-11

(b)

x	-6	-2	2	6	10
y	-11	-8	-5	-2	1

(c)

x	1	2	3	4
y	2	5	10	17

(d)

x	-3	0	4	9
y	6	6	6	6

(e) In table (b) the x-values go up in steps of 4, not 1. Explain why the first differences of the y-column are therefore *not* the slope, and state what you must divide by.

Part C — Parallel and Perpendicular Conditions

None of the given lines below is in the form $y = mx + b$, so none of them shows you its slope. Rearrange first, take the slope you need, then treat the given point exactly as in Part A.

6. Determine the equation of each line in the form $y = mx + b$.

(a) through $(4, -1)$ and parallel to $2x + 5y = 10$

(b) through $(4, -1)$ and perpendicular to $2x + 5y = 10$

(c) through $(-6, 5)$ and parallel to $3x - 4y - 12 = 0$

(d) through $(-6, 5)$ and perpendicular to $3x - 4y - 12 = 0$

(e) through $(9, -2)$ and parallel to the line joining $(1, 3)$ and $(5, -5)$

(f) through $(9, -2)$ and perpendicular to the line joining $(1, 3)$ and $(5, -5)$

7. Consider the line with equation $kx + 3y - 6 = 0$, where k is a constant.

(a) Rearrange the equation into the form $y = mx + b$, with k carried along as a letter. State m and b.

(b) Determine the value of k for which this line is **parallel** to $y = 4x - 1$.

(c) Determine the value of k for which this line is **perpendicular** to $y = 4x - 1$.

(d) Determine the value of k for which this line passes through $(5, -3)$.

(e) For which value of k is the line horizontal? State its equation for that k.

(f) Explain why *no* value of k makes this line vertical.

8. Each pair below describes two lines. Three of the pairs describe the **same** line twice. Build the $y = mx + b$ equation of each line and decide.

(a) A: through $(0, -3)$ with slope $\frac{2}{3}$.
B: $2x - 3y - 9 = 0$.

(b) C: through $(2, 7)$, parallel to $y = -x + 1$.
D: $x + y - 9 = 0$.

(c) E: through $(-1, 4)$ and $(3, -4)$.
F: through $(0, 2)$, perpendicular to $y = \frac{1}{2}x$.

(d) G: through $(1, 1)$ and $(4, 7)$.
H: $2x - y + 1 = 0$.

(e) Name the odd pair out and state exactly how those two lines are related to each other.

Part D — Horizontal and Vertical Lines

9. State the equation of each line. Three of them cannot be written in the form $y = mx + b$ — mark those with a star.

(a) through $(-5, 9)$, parallel to the x-axis

(b) through $(-5, 9)$, perpendicular to the x-axis

(c) through $(3, -2)$ and $(3, 11)$

(d) through $(-7, 4)$ and $(6, 4)$

(e) through $(0, -6)$ with slope 0

(f) with x-intercept 8 and no y-intercept

10. Consider the line through $P(-4, 1)$ and $Q(-4, -7)$.

(a) Substitute into $m = \frac{y_2 - y_1}{x_2 - x_1}$ and describe exactly what goes wrong. State the slope using the correct word.

(b) State the equation of the line.

(c) A student says "the slope is 0, so the equation is $y = -4$." Identify **both** errors in that one sentence.

(d) Explain why *no* choice of m and b can make $y = mx + b$ produce this line. Your explanation must mention how many y-values the input $x = -4$ has, and how many y-values $y = mx + b$ can return for a single x.

(e) State the domain and the range of this relation, and of the line in question 9(a).

Part E — Building from the Two Intercepts

An x-intercept is usually written as a bare number, which makes it look like a different kind of object from a point. It is not: "x-intercept 5" is the point $(5, 0)$, and once you write it that way, Part A's method finishes the job.

11. Determine the equation of each line in the form $y = mx + b$.

(a) x-intercept $(5, 0)$, y-intercept $(0, -3)$

(b) x-intercept $(-4, 0)$, y-intercept $(0, 6)$

(c) x-intercept -9, y-intercept -6

(d) x-intercept 7, y-intercept 2

(e) In every part above, one of the two given facts went straight into the equation with no work at all, while the other had to be handled as an ordinary point. Say which is which, and why.

(f) A student claims to have a line with x-intercept 4 and y-intercept 0. Explain why no line other than the x-axis can have both of those, and state what the x-intercept must be for any other line whose y-intercept is 0.

Part F — A Model Built from a Fee and a Rate

12. A tool-rental shop charges a flat booking fee of \$18.00, plus \$6.50 for every hour the tool is kept. Let C be the total cost in dollars for a rental of h hours.

(a) Write the equation relating C and h in the form $C = mh + b$.

(b) State the value of m and interpret it in a full sentence, **with units**.

(c) State the value of b and interpret it in a full sentence, **with units**.

(d) What does a 7-hour rental cost?

(e) A customer is charged \$76.50. For how many hours did they keep the tool?

(f) Parts (d) and (e) run the same equation in opposite directions. Name the operation each one needs.

(g) A rival shop charges no booking fee but \$9.00 per hour. Write its equation, then determine the rental length for which the two shops charge exactly the same.

(h) Which shop is cheaper for a 4-hour job, and by how much? Explain, in terms of m and b, why the rival wins for short rentals but loses for long ones.

Part G — Error Analysis

13. Asked for the equation of the line with slope -3 that passes through $(2, 5)$, a student writes:

$$y = mx + b \;\Rightarrow\; 5 = m(2) + (-3) \;\Rightarrow\; 2m = 8 \;\Rightarrow\; m = 4, \qquad \text{so } y = 4x - 3.$$

(a) Identify precisely what the student substituted where, and which letter should have been the unknown.

(b) Determine the correct equation.

(c) Verify that the student's line $y = 4x - 3$ really does pass through $(2, 5)$. Since it does, explain why checking the point would *not* have caught this error — and state the check that does.

14. Asked for the equation of the line through $(-3, 8)$ that is **parallel** to $y = \frac{2}{3}x - 4$, a student writes $m = -\frac{3}{2}$ and then $y = -\frac{3}{2}x + \frac{7}{2}$.

(a) Name the error in one sentence.

(b) Determine the correct equation.

(c) The student's arithmetic after the slope is flawless. State the question their answer *is* the correct answer to.

15. Asked for the equation of the line through $(6, -1)$ that is **perpendicular** to $y = \frac{4}{5}x + 2$, a student uses $m = \frac{5}{4}$ and gets $y = \frac{5}{4}x - \frac{17}{2}$.

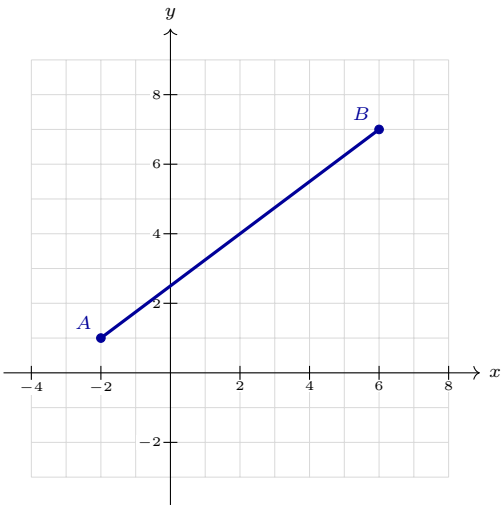
(a) Multiply the student's slope by $\frac{4}{5}$. What should the product have been, and what does that tell you?

(b) Determine the correct equation.

(c) Write a one-line check that catches this mistake every time. It must mention the *signs* of the two slopes, not just the reciprocal.

Part H — Stretch: The Perpendicular Bisector

16. The **perpendicular bisector** of a segment is the line that cuts the segment in half and meets it at 90°. It takes one ingredient from each of two units: the **midpoint** formula from Unit 6, and the **negative-reciprocal slope** plus $y = mx + b$ from this lesson in Unit 7. Consider the segment joining $A(-2, 1)$ and $B(6, 7)$, drawn below.



(a) Determine the midpoint M of AB. *(Unit 6.)*

(b) Determine the slope of AB. *(Unit 6.)*

(c) State the slope of the perpendicular bisector. *(Unit 7, this lesson.)*

(d) Use M and the slope from (c) to determine the equation of the perpendicular bisector in the form $y = mx + b$, then draw it on the grid.

(e) Verify your equation by substituting M into it.

(f) Show that $P(5, 0)$ lies on your line, then compute PA and PB using the distance formula from Unit 6. What do you notice?

(g) Your answer to (f) is a property every point of a perpendicular bisector has. State that property in one sentence, without using coordinates.

17. *(Multiple Choice)* A line passes through $(-3, 4)$ and is parallel to the line $4x + 3y - 9 = 0$. The value of the y-intercept of this line is

A. -8 B. 0 C. 8 D. $\frac{25}{4}$

18. *(Numerical Response)* The line that passes through $(-4, 7)$ and is perpendicular to $3x - 2y + 8 = 0$ intersects the line $y = 5$ at the point (p, q). The value of $p + q$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 3 — Extra Practice

The General Form of a Linear Equation

Part A — Into General Form

*The house rule for this unit: in $Ax + By + C = 0$ the numbers A, B and C are **integers** and A is **positive**. Clear every fraction first by multiplying the whole equation by the common denominator, then gather. An answer such as $-3x + y - 4 = 0$ is a true equation of the line but it is* not *in this unit's general form — multiplying through by -1 is part of the job, not an optional extra.*

1. Convert each equation to general form, $Ax + By + C = 0$, with integer A, B, C and $A > 0$.

(a) $y = 5x - 8$

(b) $y = -\frac{2}{3}x + 4$

(c) $y = \frac{3}{4}x - \frac{5}{8}$

(d) $y = -\frac{5}{2}x - \frac{7}{6}$

(e) $y = \frac{7}{3}x + \frac{1}{2}$

(f) $y = -x + \frac{9}{5}$

(g) In parts (a), (c) and (e) the slope is positive. If you gathered by leaving y where it stood and sliding the x-term across, your first line of work ended with a *negative* A. Write down what that first attempt looked like in part (a), name the one operation that repairs it, and explain why that operation cannot change which points are on the line.

(h) Parts (c) and (e) both needed a common denominator. State the denominator you used in each, and explain why multiplying *every* term by it — including the term that was already a whole number — is compulsory.

2. Determine the equation of each line in general form. None of these arrives as $y = mx + b$, so build that form first (or go straight to general form if you can see how).

(a) slope $-\frac{3}{5}$, y-intercept $(0, 7)$

(b) through $(4, -3)$ and $(10, 1)$

(c) x-intercept $(-6, 0)$, y-intercept $(0, 4)$

(d) through $(-2, 5)$ and $(6, -1)$

(e) $\frac{x}{3} + \frac{y}{7} = 1$

(f) $y - 2 = \frac{5}{6}(x + 3)$

(g) Equation (e) is written in **intercept form**. Read both intercepts straight off it, then confirm them from the general form you produced. State, in one sentence, the rule that intercept form obeys.

Part B — Out of General Form, Two Ways

*There are two routes out of $Ax + By + C = 0$. **Route 1:** solve for y, then read m and b off the result. **Route 2:** stay in general form — take $m = -\frac{A}{B}$, set $y = 0$ and solve for the x-intercept, set $x = 0$ and solve for the y-intercept. Both are correct. They are not equally fast on every question.*

3. For each line, determine the slope, the x-intercept and the y-intercept. Give both intercepts as ordered pairs. Beside each answer write **R1** or **R2** for the route you actually used.

(a) $4x - 5y + 20 = 0$

(b) $3x + 8y - 24 = 0$

(c) $7x - 2y - 9 = 0$

(d) $6x + 9y + 30 = 0$

(e) $x - 4y = 0$

(f) $5x + 3y - 2 = 0$

4. Now compare the routes.

(a) Name one part of question 3 where solving for y was clearly the faster route, and say what feature of the numbers made it so.

(b) Name one part where staying in general form was clearly faster, and say what feature of the numbers made it so.

(c) Part (e) is unusual: its two intercepts land on top of each other. State the single point, and say what $C = 0$ always tells you about a line.

(d) In part (d) of question 3, all three coefficients share a common factor. Divide it out first and redo the question. Does the line change? Explain what dividing by a common factor does and does not alter.

5. The direct route deserves formulas of its own.

(a) Starting from $Ax + By + C = 0$, substitute $y = 0$ and solve for x; then substitute $x = 0$ and solve for y. Write both intercepts in terms of A, B and C.

(b) Test your two formulas on $9x - 4y + 36 = 0$, then check both answers by substituting the points back into the original equation.

(c) What do your formulas say when $C = 0$? Does the answer agree with question 3(e)?

Part C — The Missing Coefficients: $B = 0$ and $A = 0$

General form has a variable y can stand in for and a variable x can stand in for, and nothing forces both to appear. Knock out one coefficient and the line does not disappear — it turns.

6. Two equations, one letter short each: $L_1 : \ 3x - 12 = 0$ $L_2 : \ 2y + 7 = 0$.

(a) State A, B and C for each, reading the missing term as a coefficient of 0.

(b) Solve each for the variable it contains and describe the graph in words (direction, and where it sits).

(c) State the slope of each, using the correct word where there is no number.

(d) State every intercept each line has. Each of them is missing an intercept — say which one and why.

(e) Apply $m = -\dfrac{A}{B}$ to L_1 and describe exactly what goes wrong. Then apply it to L_2 and say why it behaves.

(f) Exactly one of L_1, L_2 can be written in the form $y = mx + b$. Write it that way, and explain in one sentence why the other cannot be written that way at all.

(g) **The point of this question.** Complete and justify the sentence: "Every line in the plane can be written in _____ form, but only _____ lines can be written in $y = mx + b$ form." Your justification must name the coefficient that is allowed to be zero in general form and explain what happens if you try to let the matching thing be zero in $y = mx + b$.

(h) Write, in general form with integer coefficients and the leading non-zero coefficient positive, the vertical line through $(-5, 2)$ and the horizontal line through $(-5, 2)$.

(i) Your two answers to (h) are perpendicular. Explain why the test $m_1 m_2 = -1$ cannot be used to confirm it, and state the wording the unit uses for this exception.

Part D — Is the Point on the Line? Substitute, Do Not Graph

A point lies on a line exactly when its coordinates make the equation true. In general form the test is as clean as it ever gets: put x and y in, evaluate the left side, and see whether you land on 0. No rearranging, no graphing.

7. Line L has equation $5x - 3y - 12 = 0$. For each point, substitute, show the value the left side takes, and state **ON** or **OFF**.

(a) $(3, 1)$

(b) $(6, 5)$

(c) $(0, -4)$

(d) $(-6, -14)$

(e) $\left(\frac{12}{5}, 0\right)$

(f) $(9, 10)$

(g) Part (b) is a **near miss**. Determine the y-value L really has when $x = 6$, and state how far $(6, 5)$ sits from the line vertically.

(h) On a hand-drawn grid at six squares to the page, a one-unit miss is about the width of a pencil line. Explain why that makes graphing a poor test, and why substitution never has that problem.

(i) What does part (e) tell you about the point $\left(\frac{12}{5}, 0\right)$ specifically — not just that it is on L? Name it.

(j) Two of the six points gave the same non-zero value for the left side. Those two points lie on a *different* line that is parallel to L. Write that line's equation in general form, and explain how you got it without doing any new work.

Part E — Parallel and Perpendicular Without Rearranging

You never have to solve for y to compare two lines. The slope of $Ax+By+C=0$ is $-\dfrac{A}{B}$; read it off both lines and compare. Equal slopes mean parallel; a product of -1 means perpendicular; anything else means neither.

8. For each pair, write down $-\dfrac{A}{B}$ for both lines and classify the pair as **parallel**, **perpendicular**, or **neither**. Do not solve for y.

(a) $3x-4y+8=0$ and $9x-12y-5=0$

(b) $2x+5y-7=0$ and $5x-2y+3=0$

(c) $6x-9y+4=0$ and $4x-6y-11=0$

(d) $x+3y-6=0$ and $6x-2y+1=0$

(e) $7x+2y-9=0$ and $2x+7y+5=0$

(f) $4x-5y+2=0$ and $8x-10y+4=0$

(g) Pair (e) is the trap. The coefficients 7 and 2 are swapped from one line to the other, which looks like a reciprocal. Multiply the two slopes and state the product. What extra thing must happen before a swap makes lines perpendicular?

(h) In pair (b) the swap *did* work. Compare the signs in (b) with the signs in (e) and state the exact rule for turning $Ax+By+C=0$ into a perpendicular line.

(i) Pair (f) is labelled parallel by the slope test, but something stronger is true. Say what, and how you can see it in one glance at the coefficients.

9. **A shortcut worth proving.** Let A and B both be non-zero.

(a) Write the slope of $Ax+By+C=0$ and the slope of $Bx-Ay+D=0$, then multiply them. Conclude.

(b) Explain why $Ax+By+D=0$ is parallel to $Ax+By+C=0$ for every value of $D\neq C$, and what changing D does move.

(c) Use (b) to write, in general form, the line through $(4,-7)$ parallel to $5x+2y-3=0$.

(d) Use (a) to write, in general form, the line through $(4, -7)$ perpendicular to $5x + 2y - 3 = 0$.

(e) Verify both of your answers by checking that $(4, -7)$ satisfies them and that the two slopes come out as they should.

Part F — The Same Line Wearing a Different Coat

10. Consider $P: \ 2x - 3y + 6 = 0$ and $Q: \ -6x + 9y - 18 = 0$.

(a) Find the number k for which multiplying every term of P by k produces Q exactly.

(b) Q is not in this unit's general form. Put it there, in lowest terms, and compare with P.

(c) Confirm with slope and both intercepts that P and Q are the same line.

(d) Now take $R: \ 4x - 6y - 5 = 0$. Show that no single multiplier k turns P into R, and state how P and R *are* related.

(e) State the test in general: two equations $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$ describe the same line exactly when _____, and are merely parallel when _____.

(f) Determine k so that $kx - 12y + 24 = 0$ is the **same line** as P.

(g) Determine k so that $kx - 12y + 30 = 0$ is **parallel to but different from** P. Compare your answer with (f) and explain why the same k can do two different jobs.

(h) A student says "P and Q can't be the same line — one has $A = 2$ and the other has $A = -6$, and A is the slope." Correct both halves of that sentence.

Part G — Error Analysis

11. Asked to write $y = 4x + 7$ in general form, a student slides the x-term across and writes

$$-4x + y - 7 = 0.$$

(a) Test $(0, 7)$ and $(1, 11)$ in the student's equation. Is it a true equation of the line?

(b) So the student is not wrong about the line. State precisely which part of the *house rule* they have broken.

(c) Repair it, and state the corrected general form.

(d) Explain why multiplying an equation by -1 cannot change which points satisfy it. Your reason must mention what -1 does to the right-hand side.

12. Asked for the intercepts of $3x + 5y - 30 = 0$, a student writes: "$A = 3$ and $B = 5$, so the x-intercept is 3 and the y-intercept is 5."

(a) Determine the actual intercepts as ordered pairs.

(b) State what A and B really control between them, and name the coefficient the student ignored completely.

(c) Find the general form of the line that really does have x-intercept 3 and y-intercept 5. Compare its coefficients with the student's line and describe what has happened to A and B.

(d) Write the two correct intercept formulas from question 5(a) on one line as a rule the student could memorise, and say in one sentence why A appears in the x-intercept and B in the y-intercept, rather than the other way round.

13. Asked for the slope of $7x + 4y - 16 = 0$, a student writes $m = \dfrac{7}{4}$, taking the sizes of A and B and ignoring the signs.

(a) Solve the equation for y and state the true slope. Name the single thing the student dropped.

(b) Determine both intercepts of this line. Both are positive, so sketch the direction the line must run and state the sign the slope has to have.

(c) Write a one-line sign check that catches this error every time, using nothing but the signs of A and B.

(d) The same student gets the right answer for $7x - 4y - 16 = 0$ by the same wrong method. Explain why, and say why getting the right answer here is more dangerous than getting the wrong one in part (a).

Part H — Stretch: Solving for the Coefficients Themselves

14. The line $Ax + By + 12 = 0$ passes through $(3, 6)$ and $(-6, 0)$.

(a) Substitute each point to obtain two equations in the unknowns A and B. Notice that one of them contains only one unknown.

(b) Solve for A and B, and state the equation of the line in general form.

(c) Determine the slope and both intercepts of your line, then check that both given points satisfy the equation.

(d) Two points normally determine one line, but a general-form equation can be multiplied through by any non-zero number. Explain why fixing $C = 12$ in advance is what makes A and B come out to single values, and what would have gone wrong had the question fixed $C = 0$ instead.

(e) Suppose the question had said $Ax + By + 24 = 0$ through the same two points. Predict A and B without redoing the algebra, then verify.

15. Consider the line $8x + By - 20 = 0$, where B is unknown.

(a) Determine B so that this line is **perpendicular** to $3x + 4y - 9 = 0$. Use $-\dfrac{A}{B}$ for both slopes — do not solve either equation for y.

(b) Determine B so that this line is **parallel** to $2x - 5y + 1 = 0$.

(c) Determine B so that this line is **vertical**, and state its equation for that B.

(d) Explain why *no* value of B makes this line horizontal. Contrast this with part (c), and say which coefficient would have to change for a horizontal line to be possible.

16. *(Multiple Choice)* The general form of the line through $(0, -5)$ that is perpendicular to $4x - 6y + 9 = 0$ is

A. $3x + 2y + 10 = 0$ B. $2x - 3y - 15 = 0$ C. $3x + 2y - 10 = 0$ D. $2x + 3y + 15 = 0$

17. *(Multiple Choice)* The lines $kx + 8y - 3 = 0$ and $2x - 6y + 5 = 0$ are perpendicular. The value of k is

A. 24 B. -24 C. $\dfrac{8}{3}$ D. $-\dfrac{8}{3}$

18. *(Numerical Response)* The line with equation $5x + 2y - 40 = 0$ has x-intercept $(p, 0)$ and y-intercept $(0, q)$. The value of $p + q$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 4 — Extra Practice

Using Point-Slope Form to Determine the Equation of a Line

Part A — Substituting Before Simplifying: the Sign Traps

*Point-slope form is $y-y_1=m(x-x_1)$ — two **subtractions**. When x_1 or y_1 is negative, the subtraction of a negative becomes an addition, and that is where marks are lost. In question 1, **write the raw substitution first**, brackets and double signs and all, and only then tidy it.*

1. State the equation of each line in point-slope form. Show the raw substitution — for example $y-(-3)=4\big(x-(-5)\big)$ — and then the simplified point-slope equation.

(a) $(-5,-3)$, slope 4

(b) $(-2,7)$, slope $-\dfrac{5}{3}$

(c) $(6,-11)$, slope -1

(d) $(-9,0)$, slope $\dfrac{2}{7}$

(e) $(0,-4)$, slope -6

(f) $\left(\dfrac{1}{2},-\dfrac{5}{2}\right)$, slope 8

2. Each equation below is already in point-slope form. For each one state (i) the slope, (ii) the point (x_1,y_1) that was used, (iii) **one other point** on the line with integer or exact coordinates, found by stepping off the slope, and (iv) the y-intercept as an ordered pair.

(a) $y+6=\dfrac{3}{5}(x-10)$

(b) $y-2=-\dfrac{4}{3}(x+9)$

(c) $y-8=-\dfrac{1}{4}(x+12)$

(d) $y+\dfrac{3}{2}=2\left(x-\dfrac{1}{2}\right)$

Part B — Either Point Works

The lesson says any point on a line may be used for (x_1,y_1). Question 3 makes that claim do some work.

3. A line passes through $A(-4,9)$ and $B(6,-1)$.

(a) Determine the slope of AB. *(Do this first — point-slope form cannot start without it.)*

(b) Write the point-slope equation of the line using A as (x_1,y_1).

(c) Write the point-slope equation of the line using B as (x_1,y_1).

(d) Convert each of your two answers to general form, $Ax + By + C = 0$.

(e) Your two point-slope equations do not look alike. Explain, in one or two sentences, why they must still describe the same line.

(f) The midpoint of AB is also on the line. Find it, write a *third* point-slope equation from it, and confirm it gives the same general form.

4. Four point-slope equations are given below. Three of them describe the **same** line; one does not.

(a) $y + 2 = \frac{2}{3}(x - 3)$ (b) $y = \frac{2}{3}(x - 6)$ (c) $y + 6 = \frac{2}{3}(x + 3)$ (d) $y - 5 = \frac{2}{3}(x + 3)$

(e) Convert all four to general form and name the odd one out.

(f) Exactly one of the two ingredients — the slope or the point — differs in the odd equation. Say which, and state precisely how the odd line is related to the other three.

Part C — One Line, All Three Forms

5. A line passes through $(-8, 1)$ and $(4, -5)$.

(a) Write its equation in **point-slope form**, $y - y_1 = m(x - x_1)$.

(b) Rearrange your answer into **slope y-intercept form**, $y = mx + b$.

(c) Clear the fraction and write it in **general form**, $Ax + By + C = 0$, with A positive.

(d) Which of the three forms shows you a point on the line *and* the slope with no work at all?

(e) Which form hands you the y-intercept with no work at all?

(f) Which form has no fractions anywhere, and which form is the only one of the three that a **vertical** line can be written in? Explain the second answer.

Part D — Indirect Descriptions: Decode First, Substitute Second

None of the lines in questions 6 and 7 hands you a slope and a point. Each description hides one or both. Write down the slope and the point you extracted before you touch the formula.

6. Determine the equation of each line in general form, $Ax + By + C = 0$.

(a) through the midpoint of $A(-3, 10)$ and $B(9, -2)$, with slope -2

(b) through $(3, -1)$, perpendicular to the line through $(0, 4)$ and $(6, 0)$

(c) through the x-intercept of $5x - 2y - 20 = 0$, parallel to $y = \frac{3}{4}x$

7. Determine the equation of each line in general form, $Ax + By + C = 0$.

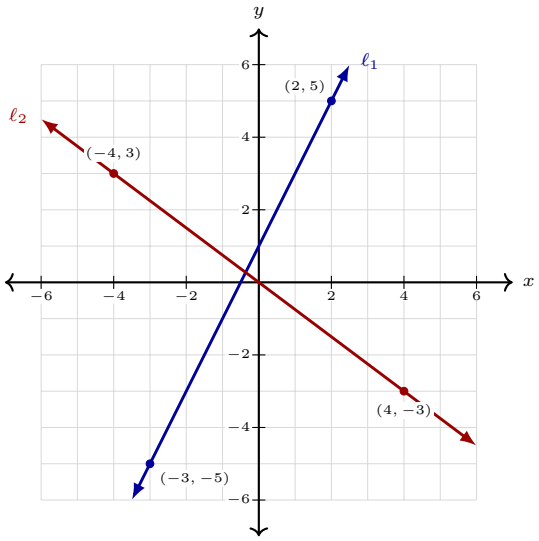
(a) through the y-intercept of $3x - 7y + 21 = 0$, perpendicular to $2x + 5y - 15 = 0$

(b) through the point on $y = 4x + 7$ whose x-coordinate is -2, parallel to $6x + 9y - 18 = 0$

(c) through $(-5, 2)$, perpendicular to the line joining the origin to the midpoint of $C(1, -4)$ and $D(7, 8)$

Part E — From a Grid

8. Two lines are drawn on the grid. Each passes through the two marked lattice points.



(a) For ℓ_1: write *two* point-slope equations, one from each marked point, then give the general form.

(b) For ℓ_2: do the same.

(c) The two lines clearly cross. Use the product of their slopes to decide whether they cross at 90°, and state the product.

Part F — Two Data Points, One Model

9. An appliance technician charges a fixed call-out fee plus a constant hourly rate. A job that took 2 hours was billed at \$185, and a job that took 5 hours was billed at \$320. Let t be the time in hours and C the bill in dollars.

(a) Write the two data points as ordered pairs (t, C).

(b) Determine the rate of change and state what it means in a sentence, **with units**.

(c) Use point-slope form to write the model, then rearrange it to $C = mt + b$.

(d) Interpret the value of b in the context. What is it called on the invoice?

(e) Predict the bill for a job lasting 7.5 hours.

(f) A customer was billed \$500. How long did the job take?

10. In a deep lake the water temperature falls at a constant rate with depth over the first 40 metres. At a depth of 10 m the temperature is 18°C; at a depth of 25 m it is 10.5°C. Let d be the depth in metres and T the temperature in °C.

(a) Determine the rate of change and state its meaning in a sentence, **with units**.

(b) Use point-slope form to build the model, then write it as $T = md + b$.

(c) Interpret b in the context.

(d) Predict the temperature at a depth of 40 m.

(e) At what depth does the model predict 0°C?

(f) The lake is 120 m deep. Explain why the model should not be used at $d = 120$.

Part G — Error Analysis

11. Asked for the equation, in general form, of the line through $(4, -7)$ with slope -3, a student writes

$$y + y_1 = m(x + x_1) \;\Rightarrow\; y + (-7) = -3(x + 4) \;\Rightarrow\; y - 7 = -3x - 12 \;\Rightarrow\; 3x + y + 5 = 0.$$

(a) State the correct form and name the student's error in one sentence.

(b) Determine the correct equation in general form.

(c) Substitute $(4, -7)$ into the student's equation to show it fails. Then find the point on the student's line with $x = 4$.

12. Asked for the equation, in general form, of the line through $(2, 9)$ with slope $\frac{1}{2}$, a student writes $y - 2 = \frac{1}{2}(x - 9)$.

(a) Name the error precisely — say which number belongs in which slot.

(b) Determine the correct equation in general form.

(c) Convert the student's equation to general form as well. The two lines are not the same, yet they are related. State the relationship and the one point the student's line *does* pass through.

13. Asked for the equation, in general form, of the line through $(-3, 8)$ with slope $-\frac{5}{2}$, a student writes

$$y - 8 = -\tfrac{5}{2}(x + 3) \;\Rightarrow\; 2y - 16 = -5x - 15 \;\Rightarrow\; -5x - 2y + 1 = 0.$$

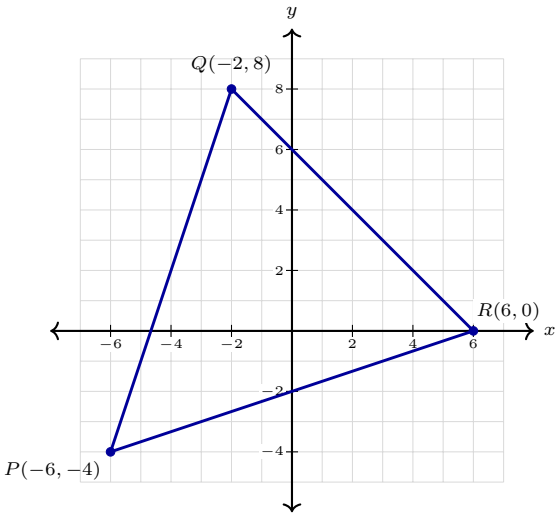
(a) Every line of the student's work is algebraically correct. Verify that $(-3, 8)$ satisfies their final equation.

(b) State the unit's house rule for general form, then write the corrected answer.

(c) Explain why the correction is a matter of *convention* rather than of truth, and why a convention is still worth following.

Part H — Stretch: The Altitudes of a Triangle

14. An **altitude** of a triangle is the line drawn from a vertex perpendicular to the opposite side. Building one needs three steps in order: the slope of the opposite side, its negative reciprocal, and then point-slope form using the vertex. Triangle PQR has vertices $P(-6, -4)$, $Q(-2, 8)$ and $R(6, 0)$.



(a) Determine the slope of each of the three sides PQ, QR and PR.

(b) Determine the equation, in general form, of the altitude from P to side QR.

(c) Determine the equation, in general form, of the altitude from Q to side PR.

(d) Determine the equation, in general form, of the altitude from R to side PQ.

(e) Verify that the point $(0, 2)$ satisfies all three altitude equations, and draw the three altitudes on the diagram.

(f) The three altitudes of *any* triangle meet at a single point. Explain why, once you had found two of the altitudes, you did not strictly need the third one to locate that point.

Part I — Multiple Choice and Numerical Response

15. *(Multiple Choice)* The equation, in general form, of the line through $(-7, 3)$ with slope $\frac{2}{5}$ is
A. $2x - 5y + 29 = 0$ B. $2x + 5y - 1 = 0$ C. $2x - 5y - 29 = 0$ D. $5x - 2y + 41 = 0$

16. *(Multiple Choice)* Which equation is equivalent to $y + 4 = -\frac{3}{8}(x - 16)$?
A. $3x - 8y - 16 = 0$ B. $3x + 8y + 16 = 0$ C. $3x + 8y - 16 = 0$ D. $3x - 8y + 64 = 0$

17. *(Numerical Response)* The line through $(-4, 5)$ perpendicular to $3x+2y-7 = 0$ is written in general form $Ax+By+C = 0$, where A, B and C are integers with A positive and no common factor greater than 1. The value of $A+B+C$ is _____. *(Record your answer as a three-digit number.)*

Lesson 5 — Extra Practice

Graphing Linear Equations: A Strategy for Every Form

Part A — Choose the Strategy Before You Draw

*Lesson 5 gives you three hand-graphing strategies: read m and b off **slope y-intercept form**; substitute $y = 0$ and $x = 0$ to get **the two intercepts**; or build a **table of values** when neither of the first two is clean. The point of this set is not that you can graph a line — it is that you pick the strategy that gets you there in the fewest steps, and can say why.*

1. For each equation, name the strategy you would use — *slope y-intercept*, *two intercepts*, or *table of values* — and give the one piece of evidence from the equation that decided it. **Do not graph anything.**

(a) $y = -\frac{3}{4}x + 5$

(b) $5x + 2y - 20 = 0$

(c) $7x - 3y - 21 = 0$

(d) $y - 4 = \frac{2}{5}(x + 5)$

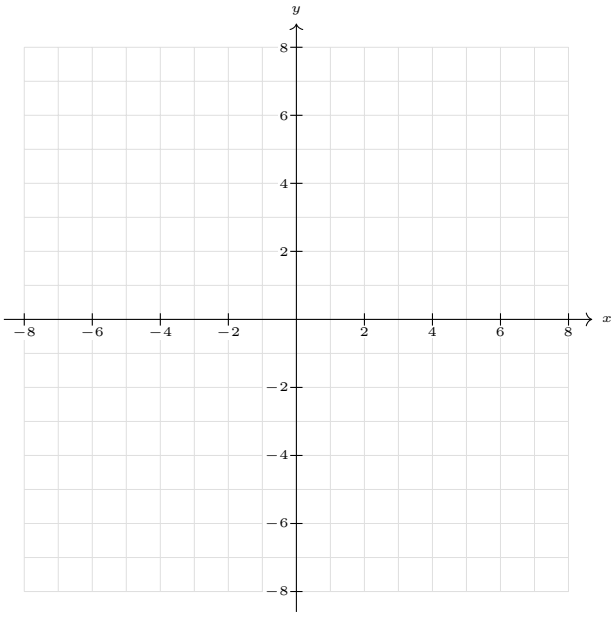
(e) $3x - 5y - 4 = 0$

(f) $y = \frac{5}{2}x$

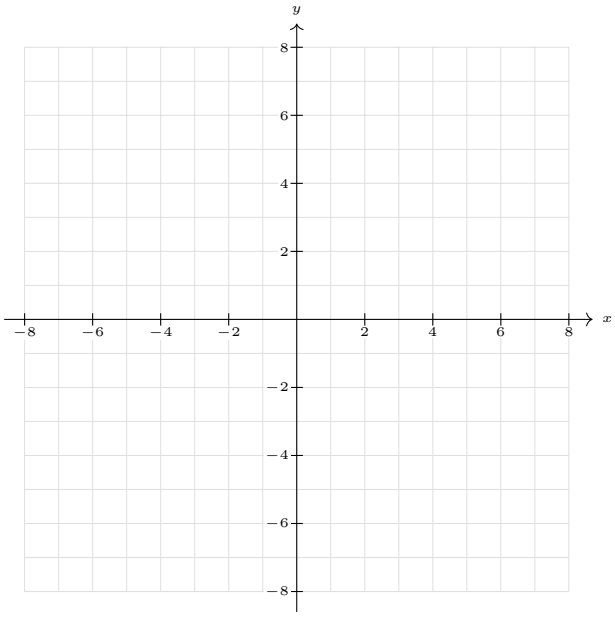
(g) Three of the six are in general form, but you should not have chosen the same strategy for all three. Say which two are two-intercept questions, and what makes the third one not.

2. **The wrong strategy is slower — prove it.** Graph each line *twice*: once by the strategy named in the left column and once by the strategy named in the right column. Show the arithmetic each time, then say which was faster and by how many steps.

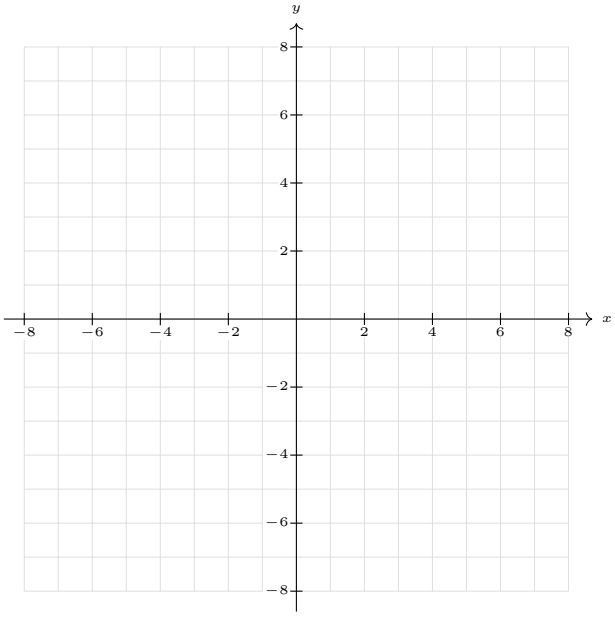
(a) $7x + 4y - 28 = 0$ *by the two intercepts*



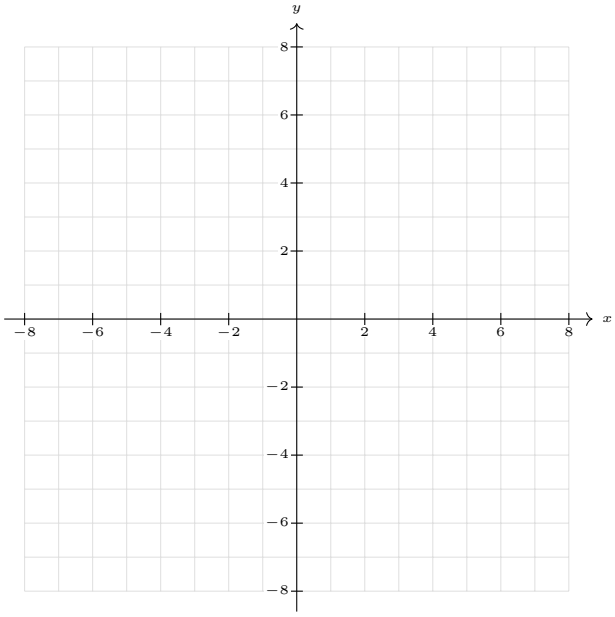
(a) $7x + 4y - 28 = 0$ *by solving for y first*



(b) $y = 3x + 2$ *by slope and y-intercept*



(b) $y = 3x + 2$ *by the two intercepts*



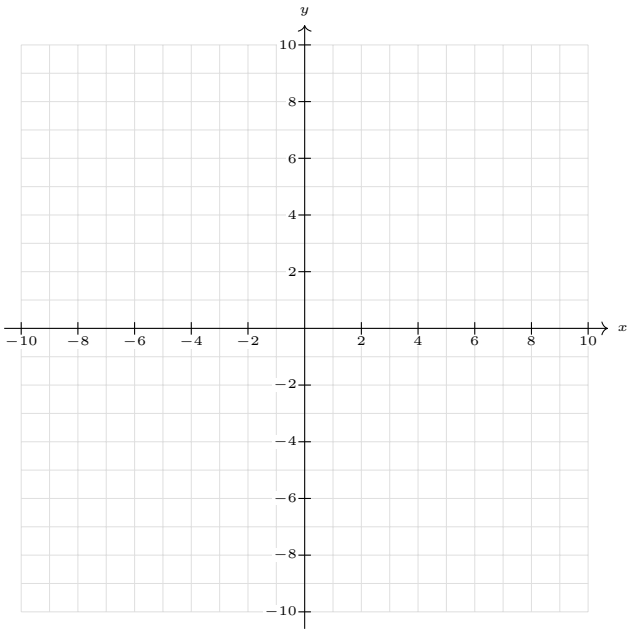
(c) In (a) the slower method produced an awkward number; in (b) the slower method produced an awkward number too — but a different *kind* of awkward. Describe both, and state the rule of thumb they suggest for reading a general-form equation.

Part B — One Strategy at a Time, on the Grid

3. Each equation is in general form and both intercepts are whole numbers. Find them by substitution, state them **as ordered pairs**, then draw the line through them.

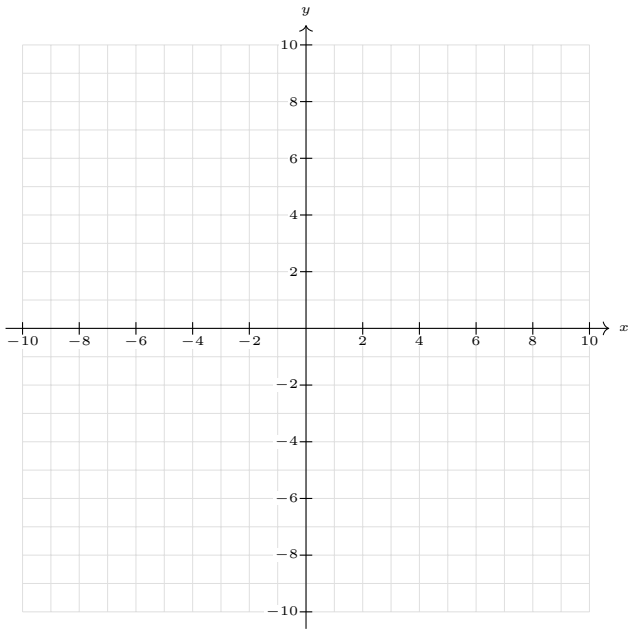
(a) $4x - 5y + 20 = 0$

x-int _____ y-int _____



(b) $3x + 8y - 24 = 0$

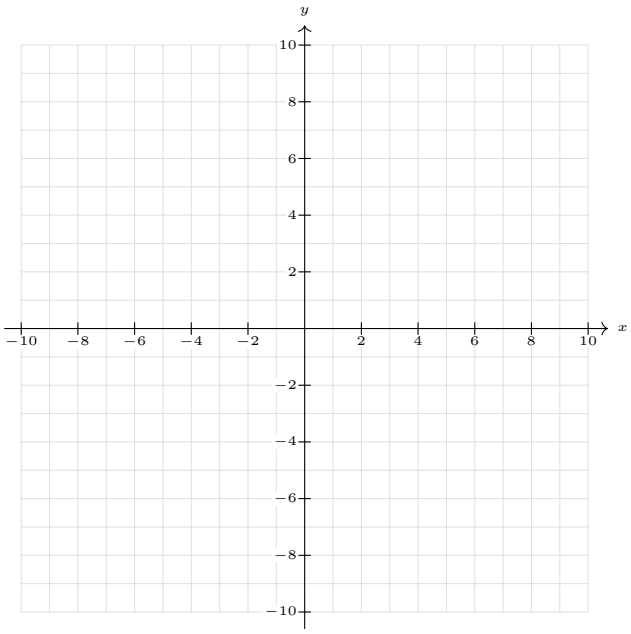
x-int _____ y-int _____



4. Each equation is in **point–slope form**. Do *not* convert it. State the point the equation hands you and the slope, then step the slope to plot at least three points and draw the line.

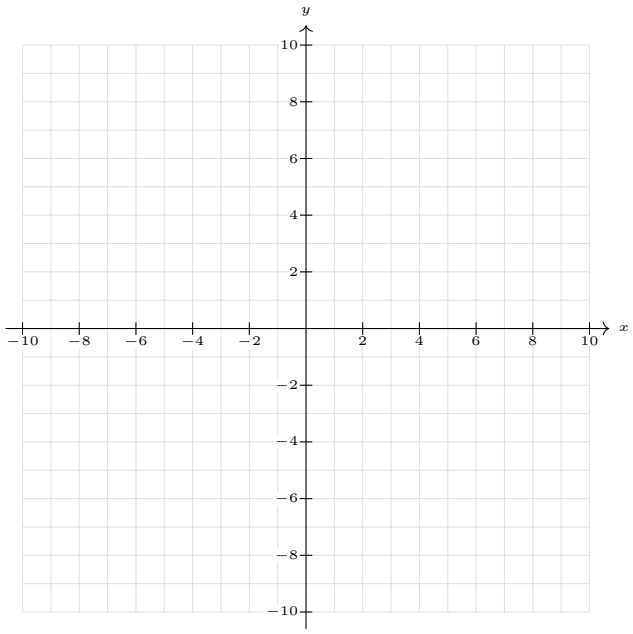
(a) $y + 2 = \frac{3}{4}(x - 4)$

point _____ $m =$ _____



(b) $y - 5 = -\frac{2}{3}(x + 3)$

point _____ $m =$ _____

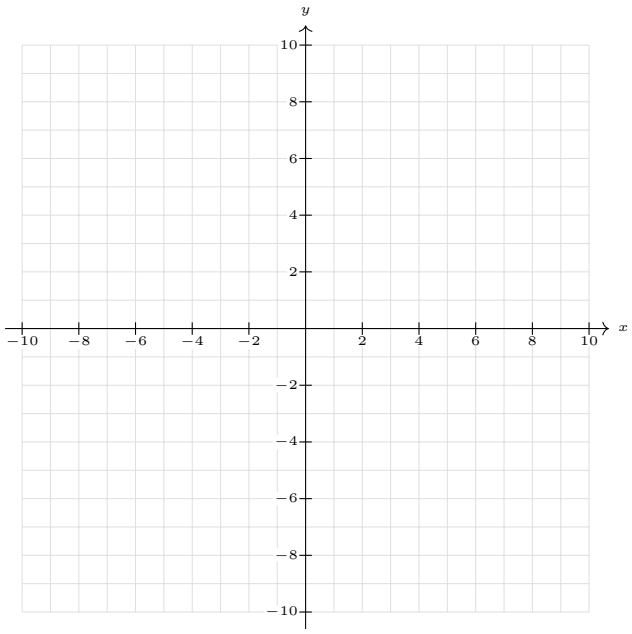


(c) Neither point the equation hands you in (a) or (b) is an intercept. Explain why point–slope form almost never hands you one, and why that costs you nothing when you are graphing.

5. **The fallback.** For $3x - 5y - 4 = 0$ the slope is $\frac{3}{5}$, the x-intercept is $\frac{4}{3}$ and the y-intercept is $-\frac{4}{5}$ — nothing is pleasant. Use a table of values instead.

(a) Complete the table, then plot the points and draw the line.

x	y	(x, y)
-7		
-2		
3		
8		



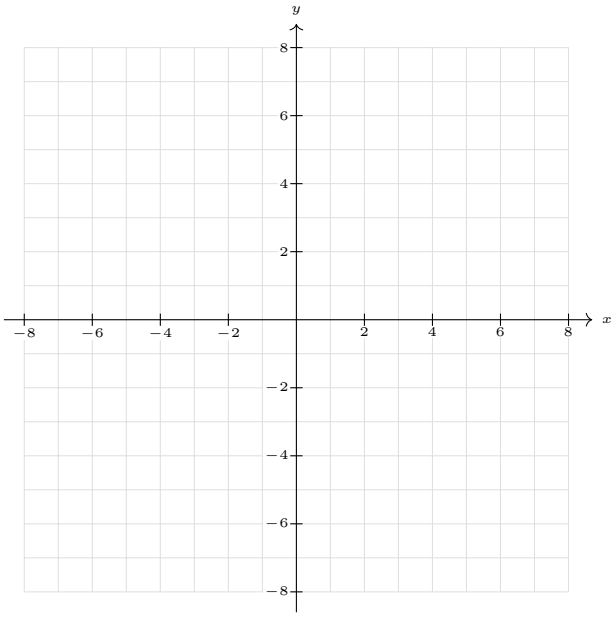
(b) The x-values step by 5, not by 1. Show, from the equation, why a step of 5 is the smallest step that keeps every y-value a whole number.

(c) A classmate says "a table of values always works, so just always use a table." Give the strongest argument against this, using question 3 as your evidence.

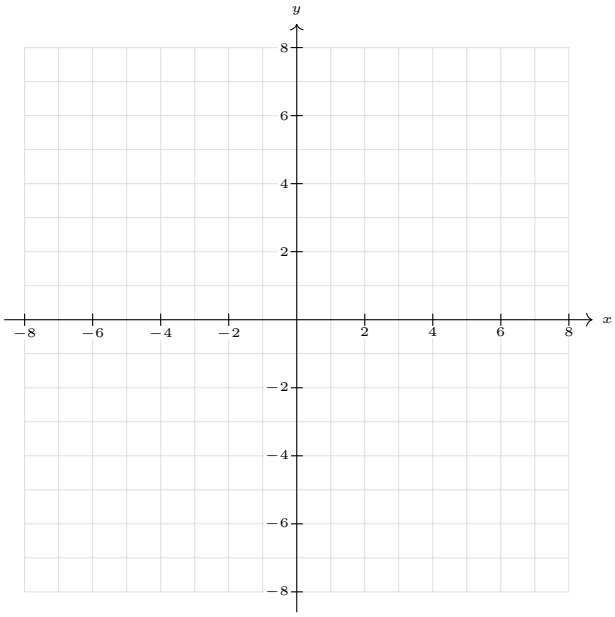
Part C — Lines With Only One Intercept

6. For each line, first say what the two-intercept strategy produces — two points, one point, or nothing usable — and then graph the line by whatever means actually works.

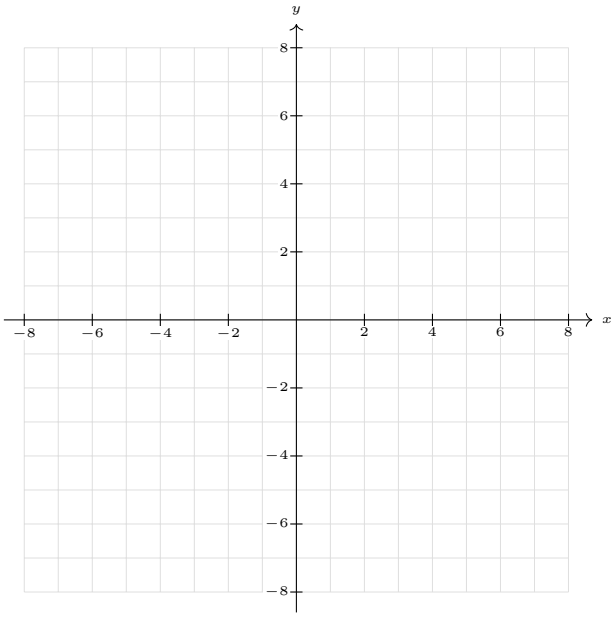
(a) $3y + 15 = 0$



(b) $2x + 7 = 0$



(c) $y = \frac{3}{2}x$



(d) One of the three has a slope of 0, one has an undefined slope, and one has a slope you can read straight off. Match them up.

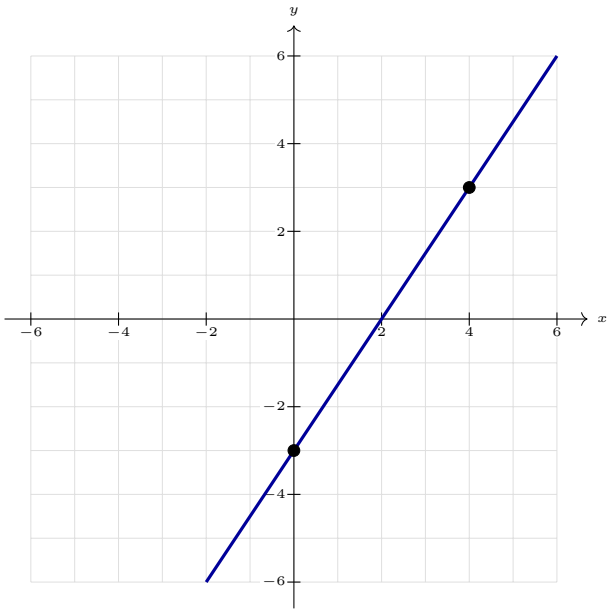
(e) In (c) the intercept strategy gives you a single point. State exactly what you must do next, and name one point you could use.

(f) Only one of the three cannot be written in the form $y = mx + b$ at all. Say which, and why the form breaks down for it.

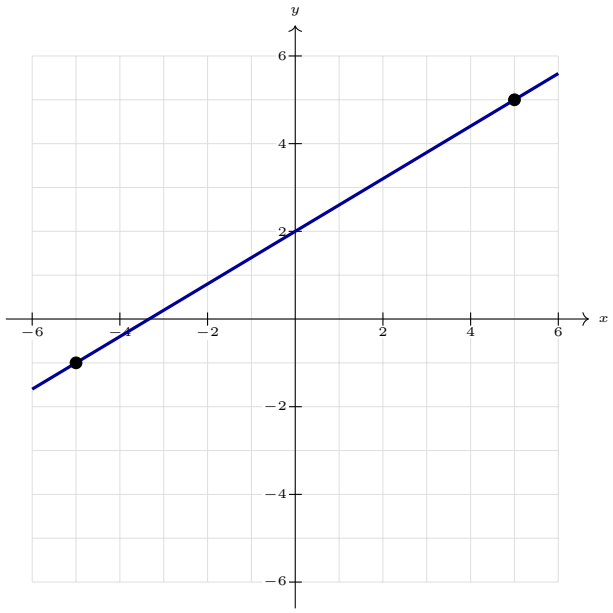
Part D — Backwards: From a Drawn Line to All Three Forms

7. Each line below is drawn through the marked lattice points. For each, write its equation in **slope y-intercept form**, in **general form** (A, B, C integers with $A > 0$), and in **point–slope form** using one of the marked points.

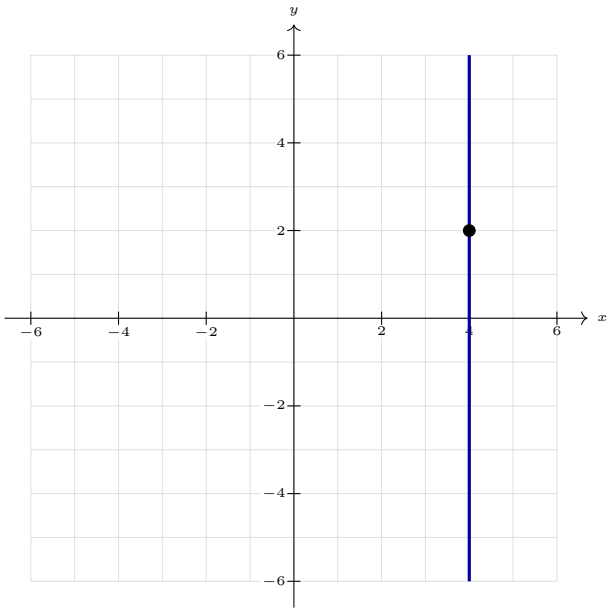
(a)



(b)



(c)



(d) One of the three lines above defeats two of the three forms. Name the line, name the two forms it defeats, and explain in one sentence what about its slope is responsible.

(e) In (b), which of the three forms would you rather have been handed if your only job was to graph the line? Why?

Part E — Recovering an Equation From a Table

8. The table below comes from a linear relation. Notice that the x-values step by 4 and that $x = 0$ is not in the table.

x	4	8	12	16	20
y	2	7	12	17	22

(a) A student reads the y-column, sees it going up by 5, and writes $m = 5$. Explain the error, then determine the correct slope.

(b) The value of b is not in the table. Determine it, showing the substitution you used.

(c) Write the relation in all three forms.

(d) Determine y when $x = 40$.

(e) Determine x when $y = 97$.

(f) Which form did you use in (d), and which in (e)? Say why they were not the same.

9. A hot-air balloon begins a steady descent. The table records the balloon's altitude A, in metres, after t minutes of descending.

t (min)	3	6	9	12	15
A (m)	1290	1230	1170	1110	1050

(a) Determine the rate of change. **Give it with units and write a sentence saying what it means for the balloon.**

(b) Write the relation in the form $A = mt + b$, and then in general form.

(c) State the A-intercept and say what it represents, in metres.

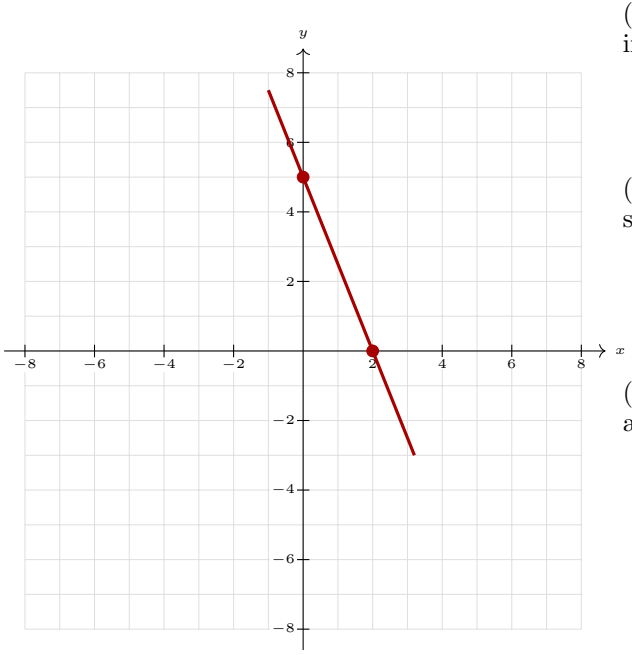
(d) Determine the t-intercept and say what event it marks.

(e) Find the altitude at $t = 40$ min, and the time at which the altitude is 600 m.

(f) You are asked to graph this relation by hand. The two intercepts are $(0, 1350)$ and $(67.5, 0)$. Explain why they are still the right two points to use, and state the scale you would put on each axis so that both fit on a grid twenty squares wide and twenty squares tall.

Part F — Error Analysis

10. Asked to graph $2x + 5y - 10 = 0$ by the two-intercept strategy, a student correctly finds that the x-intercept has value 5 and the y-intercept has value 2 — and then plots the points $(0, 5)$ and $(2, 0)$. Their line is drawn below.



(a) Name the mistake in one sentence. Which axis does each intercept belong on?

(b) Determine the equation, in general form, of the line the student actually drew.

(c) Draw the *correct* line for $2x + 5y - 10 = 0$ on the same grid, and label its two intercepts as ordered pairs.

(d) A single substitution into $2x + 5y - 10 = 0$ would have caught this before the line was ruled. State it and carry it out.

(e) Write a memory rule that names the *coordinate that is zero* at each intercept, rather than the letter in the intercept's name.

11. Asked to graph $2x - 3y - 12 = 0$, a student writes: "It's in general form, so the intercepts are the numbers in front of x and y: the x-intercept is 2 and the y-intercept is -3," and draws the line through $(2, 0)$ and $(0, -3)$.

(a) Determine the two intercepts correctly, showing both substitutions, and state them as ordered pairs.

(b) Determine the equation, in general form, of the line the student actually drew. Verify that it is not the same line.

(c) The student's shortcut is not pure invention — there *is* a way of writing a linear equation in which the intercepts really do sit in the equation, ready to read. Write $2x - 3y - 12 = 0$ in the form $\frac{x}{a} + \frac{y}{b} = 1$ and identify a and b.

(d) Using your answer to (c), state precisely what the student would have had to do to $2x - 3y - 12 = 0$ first for their shortcut to be legitimate.

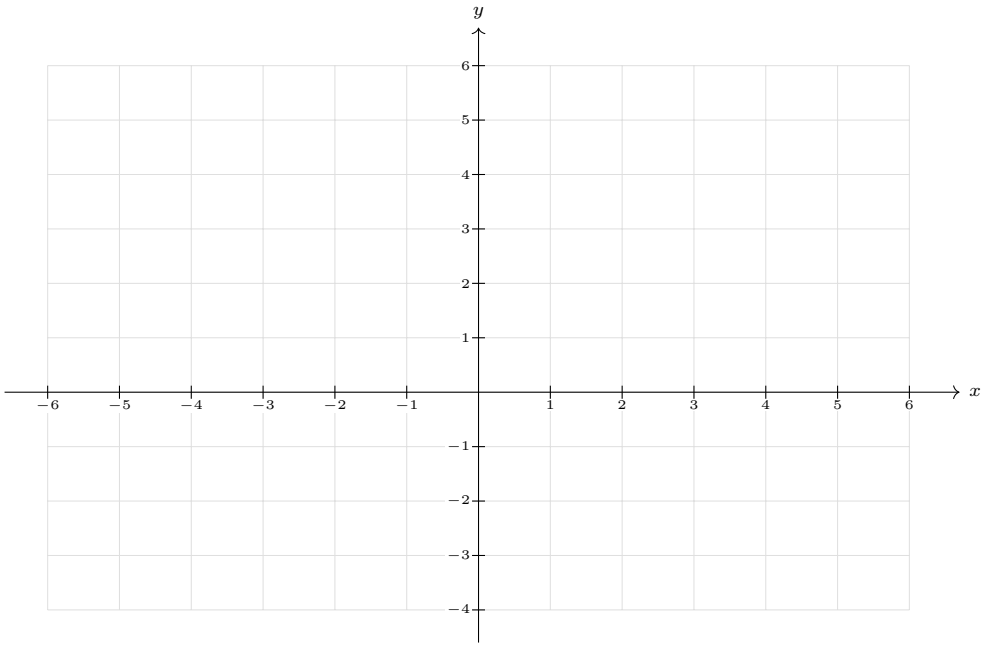
Part G — Stretch: Three Lines, One Triangle

12. The three lines below are given in three different forms.

L1: $x - y + 2 = 0$ **L2:** $y - 4 = -2(x - 2)$ **L3:** $y = -2$

(a) State the strategy you would use for each line, and why the form it is written in chose that strategy for you.

(b) Graph all three lines on the single grid below and label each line L1, L2, L3. The three lines bound a triangle.



(c) Determine the coordinates of all three vertices of the triangle **algebraically** — by solving pairs of the equations, not by reading the picture. Show which pair of lines gives each vertex, then check each vertex against your graph.

(d) One side of the triangle is horizontal. Use it as the base to determine the area of the triangle, in square units.

(e) Two of the three vertices were easy to find and one took real work. Say which, and explain what it is about L3 that made the other two easy.

Part H — Multiple Choice and Numerical Response

13. *(Multiple Choice)* Which line **cannot** be graphed by plotting its two intercepts and joining them?

A. $3x - 4y - 12 = 0$ B. $y = 2x - 6$ C. $y = -\frac{3}{5}x$ D. $x + y - 5 = 0$

14. *(Multiple Choice)* The fastest hand-graphing strategy for $9x - 4y - 36 = 0$ is to

A. solve for y and step the slope from $(0, -9)$ B. plot $(4, 0)$ and $(0, -9)$ and join them
C. build a table of values for $x = 0, 1, 2, 3$ D. plot $(9, 0)$ and $(0, -4)$ and join them

15. *(Numerical Response)* The line $3x - 8y - 24 = 0$ has x-intercept of value p and y-intercept of value q. The value of $p - q$ is _____. *(Record your answer in the numerical response box from left to right.)*

16. Write the flowchart you would follow on a test, in three sentences, starting from "Look at the form the equation is in." Your flowchart must send a reader to the right strategy for every equation in this set — including $y = \frac{5}{2}x$, $x = 4$, and $3x - 5y - 4 = 0$.

Lesson 6 — Extra Practice

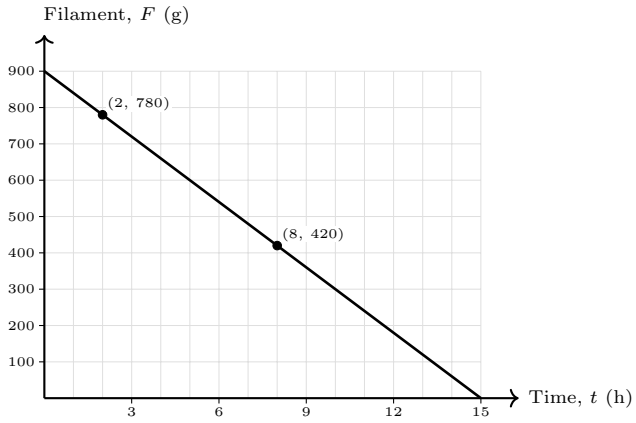
Slope in Context: Interpreting Rates of Change

Read this before you start. *Every question that asks what a slope or an intercept* means *must be answered with a* ***sentence that carries units*** *— "the fuel is decreasing at 60 grams per hour" — not with a bare number. A number alone earns no marks on an interpretation question, no matter how correct the arithmetic was.*

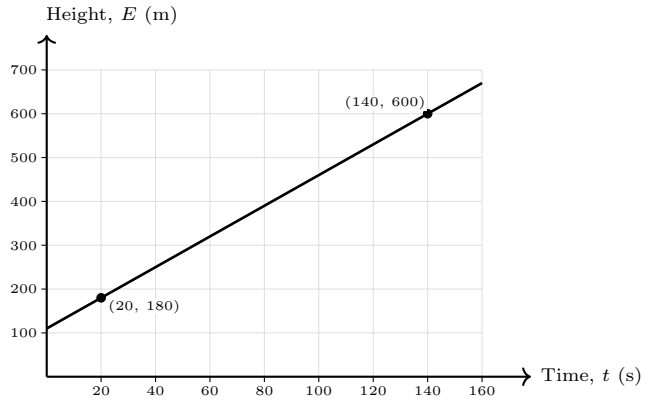
Part A — Slope Carries Units: Reading a Rate off a Graph

1. Each graph shows a linear relationship between two real quantities. For each one, calculate the slope using $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, build the units out of the two axis labels, and write a full sentence interpreting the rate.

(a) Filament remaining on a 3-D printer spool, as a function of printing time.



(b) Height above the base station of a chairlift, as a function of time. **Read the tick numbers carefully — the two axes are not on the same scale.**



(c) Interpret the slope of graph (a) in a sentence with units.
Then interpret the F-intercept of graph (a) in a sentence with units.

(d) Interpret the slope of graph (b) in a sentence with units.
Then interpret the E-intercept of graph (b) in a sentence with units.

2. **Counting squares is not the same as counting units.** A student reads graph **1(b)** by counting grid squares: "from $(20, 180)$ to $(140, 600)$ I go right 6 squares and up 4.2 squares, so the slope is $\dfrac{4.2}{6} = 0.7$."

(a) Explain exactly what the student's 0.7 measures, and why it is not the rate of change.

(b) Find the number you must multiply 0.7 by to repair the answer, and say where that number comes from.

(c) State the rule in one sentence: when may you count squares, and when may you not?

Part B — What the y-Intercept Means

3. For each model, interpret **both** the slope and the vertical intercept in sentences carrying units. Name the quantity, say whether it is rising or falling, and attach the units built from the two variables.

(a) $W = -1.6t + 40$, where W is the rinse water left in a dishwasher's reservoir in litres and t is the running time in minutes.

(b) $T = 145h + 22$, where T is the interior temperature of a pottery kiln in °C and h is the number of hours since firing began.

(c) $C = 0.92d + 55$, where C is the cost in dollars of a courier delivery of d kilometres.

4. **One of these intercepts means nothing.** Three of the four models below have a vertical intercept that describes something real. In the fourth, substituting 0 for the independent variable takes you outside the practical domain, so the intercept is a number the algebra produces but the situation cannot use.

(a) $V = -12t + 540$: V is the water in a livestock trough in litres, t hours after it was filled.

(b) $H = 2.9f + 78.5$: a forensic estimate of an adult's height H in centimetres from the length f of the forearm in centimetres. The model is used only for $20 \le f \le 32$.

(c) $P = 18n + 240$: P is the total cost in dollars of a school ski trip for n students.

(d) $M = -0.35d + 14.2$: M is the mass in kilograms of a wheel of cheese d days into its ageing.

(e) State which intercept is meaningless in context, and explain in a sentence why the situation cannot use it.

(f) For each of the other three, write the sentence with units that the intercept deserves.

Part C — Base Plus Commission: Building and Comparing Models

5. Jonah sells refurbished e-bikes and is paid a weekly base salary plus a commission on his sales. One week his sales totalled \$6 200 and he earned \$713. The next week his sales totalled \$9 000 and he earned \$825.

(a) Determine the equation of the model in the form $E = mS + b$, where E is weekly earnings in dollars and S is weekly sales in dollars.

(b) Interpret m in a sentence with units, then state Jonah's commission rate as a percent.

(c) Interpret b in a sentence with units. What would Jonah earn in a week with no sales at all?

(d) Jonah wants to earn \$1 000 in a week. What must his sales be?

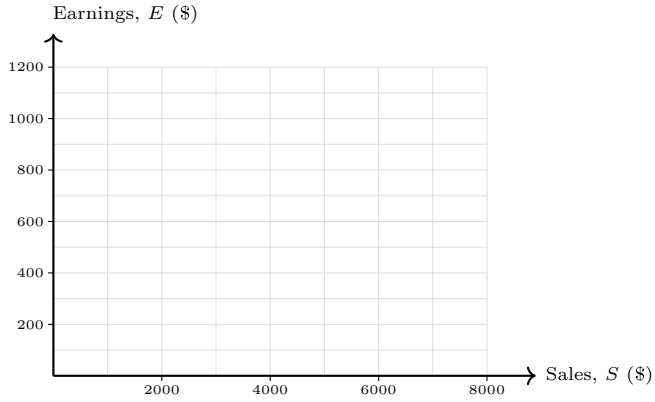
(e) Rewrite your equation from (a) in the unit's **general form** $Ax + By + C = 0$, using S for x and E for y. Remember: A, B and C must be integers and A must be positive.

6. A paddling-gear shop offers its sales staff a choice of two pay plans.

Plan A: \$520 per week, plus 4% commission on weekly sales.
Plan B: \$280 per week, plus 10% commission on weekly sales.

(a) Write an equation for each plan in the form $E = mS + b$, then graph both lines on the grid. Label each line.

(b) Interpret the slope of each plan in a sentence with units, and the intercept of each plan in a sentence with units.



(c) Determine algebraically the level of weekly sales at which the two plans pay exactly the same, and the amount paid there.

(d) Which plan pays more on sales of \$2 500, and by how much? Which pays more on sales of \$7 000, and by how much?

(e) Complete: Plan A is better whenever weekly sales are _____ \$_____, and Plan B is better whenever they are _____ \$_____.

7. The same shop adds a third option. **Plan C:** \$340 per week plus a commission rate of r, where r is chosen so that on weekly sales of \$5 000 Plan C pays exactly what Plan A pays.

(a) Determine r, and state it as a percent.

(b) On sales of \$8 000, which of Plan A and Plan C pays more, and by how much?

(c) Plan C has a smaller base than Plan A but a larger commission rate. Explain, using the two parameters, why that guarantees the answer to (b) without any further arithmetic.

Part D — Negative Rates, and What "Steeper" Means

8. Each model below has a negative rate of change. Interpret the slope of each in a sentence with units, using the right verb for the situation — *draining*, *cooling*, *depreciating*, *falling*.

(a) $V = -1450a + 11\,600$, where V is the resale value of a snowmobile in dollars and a is its age in years.

(b) $T = -1.8h + 21$, where T is the temperature in a greenhouse in °C and h is the number of hours since sunset.

(c) $L = -0.045d + 12.6$, where L is the level of a reservoir in metres and d is the number of days into a drought.

(d) At what age does model (a) predict the snowmobile is worth \$0? Is the model believable that far out? Explain.

(e) Restate the rate in (c) in **centimetres per day**. Does the reservoir fall faster in this new wording? Explain.

(f) Can you tell which of the three situations would be graphed with the steepest line? Explain why comparing -1450, -1.8 and -0.045 cannot settle it.

9. Four lines have slopes $m_1 = -4$, $m_2 = 2$, $m_3 = -\dfrac{1}{2}$ and $m_4 = 3.5$.

(a) List the four slopes in order from **least steep to steepest**.

(b) List the four slopes in order from **smallest value to largest value**.

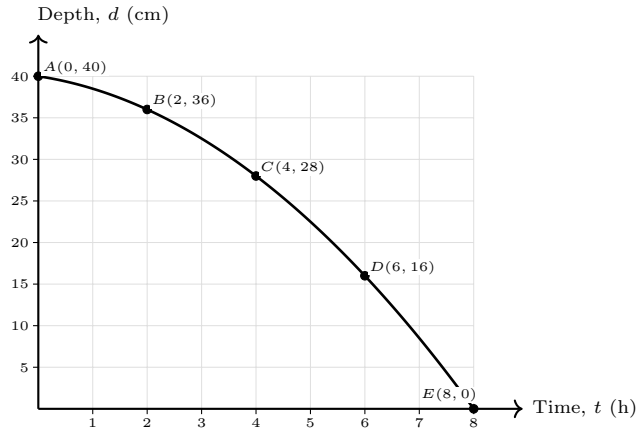
(c) The two lists are not the same. Explain in one sentence what quantity "steepness" actually compares.

(d) Tank X drains at 4 litres per minute. Tank Y fills at 2 litres per minute. Graph volume against time for each. Which graph is steeper? Which tank's graph has the *larger* slope? Answer both, and notice they are different tanks.

(e) Complete and justify: "Making a rate of change steeper makes its slope a _____ number when the rate is positive, but a _____ number when the rate is negative."

Part E — Average Rate of Change Against a Constant Rate

10. The curve shows the depth of snow on a flat roof through a spring thaw, where t is the number of hours after 8 a.m. and d is the depth in centimetres. The five marked points lie exactly on the curve.



(a) Calculate the average rate of change from A to B, and interpret it in a sentence with units.

(b) Calculate the average rate of change from C to D, and from D to E.

(c) Calculate the average rate of change for the whole thaw, A to E, and interpret it in a sentence with units.

(d) Your answers to (a), (b) and (c) are all different. Explain what that tells you about this roof, and why the phrase "the rate of change" is not enough here without naming an interval.

(e) At what time of day did the snow disappear, and during which two-hour interval was it melting fastest?

(f) A second roof also starts at 40 cm at 8 a.m. and also clears at 4 p.m., but its snow melts at a **constant** rate. Write its equation in the form $d = mt + b$ and state the rate with units.

(g) Explain why every pair of points on the second roof's graph gives the same average rate of change, while on the first roof they do not. Which roof had more snow on it at noon, and by how much?

11. Two relations are given as tables.

Table 1: depth y of a settling pond in metres, x days after treatment began.

x	0	3	6	9	12
y	17.0	11.6	6.2	0.8	−4.6

Table 2: subscribers A to a podcast, t years after launch.

t	0	1	2	3	4
A	5000	6000	7200	8640	10 368

(a) For each table, calculate the rate of change across all four successive intervals. Which relation is linear?

(b) For the linear one, write the equation and interpret both parameters in sentences with units. Then say what a negative depth means and where the model must stop.

(c) For the non-linear one, calculate the average rate of change over the whole four years and interpret it in a sentence with units. Does any single year actually grow at that rate?

Part F — Changing the Units of a Rate

12. Convert each rate to the requested units. The situation does not change — only the numbers used to describe it do.

(a) A conveyor delivers gravel at 54 tonnes per hour. Express this in tonnes per minute.

(b) Bulk cashews sell for \$23.40 per kilogram. Express this in cents per gram.

(c) A water main leaks 7.2 L per minute. Express this in litres per second, then in millilitres per second.

(d) A drone climbs 96 m per minute. Express this in metres per second, then in kilometres per hour.

(e) The drone's rate can be written as 1.6 or as 5.76. Does the drone climb faster when the larger number is used? Explain what changing the units does and does not do.

(f) Most of the six numbers you produced above are smaller than the rate you started from, but at least one is larger. State the rule that decides which way a converted rate moves.

13. A commercial ice machine is modelled by $I = 4.8t + 15$, where I is the mass of ice in the bin in **kilograms** and t is the time in **hours** since the machine was switched on.

(a) Rewrite the model as $J = mn + b$, where J is the mass of ice in **grams** and n is the time in **minutes**.

(b) State the slope of each version with its units, and interpret each in a sentence.

(c) The slope changed from 4.8 to a much larger number. Explain why the machine is not producing ice any faster.

(d) Use **both** versions to find the mass of ice after 2.5 hours, and confirm the two answers agree.

(e) Interpret the intercept of each version in a sentence with units, and explain why those two sentences say the same thing.

Part G — Error Analysis

14. Asked to interpret the slope of a graph of *litres of maple sap collected* against *hours of tapping*, a student writes:

"The slope is 1.5."

(a) The arithmetic is correct. Say precisely what the answer is missing and why a bare number cannot answer an interpretation question.

(b) Build the units from the two axis labels and rewrite the answer as a proper sentence.

15. Comparing a cooling curve of slope -4 with a heating curve of slope 2, a student writes: "-4 is less than 2, so the cooling graph is less steep."

(a) Identify the confusion in one sentence.

(b) Which graph really is steeper, and by what measure?

(c) Write a corrected statement that gets both the comparison of *value* and the comparison of *steepness* right.

16. A snow-clearing contract is modelled by $C = 85w + 320$, where C is the total cost in dollars for w clearings of a driveway over the season. A student writes: "The slope means the job costs \$85."

(a) Explain what the student has confused a *rate* with, and rewrite the interpretation correctly.

(b) Interpret the 320 in a sentence with units.

(c) Calculate the total cost of a contract with 6 clearings, and show that neither \$85 nor \$320 is that total.

Part H — Stretch and Test-Style

17. **Two models, two forms, two sets of units.** Reservoir P is filling according to $7x - 2y + 18 = 0$, where x is the number of weeks since monitoring began and y is the volume in megalitres. Reservoir Q is filling at a steady 520 000 litres per day. (1 ML = 1 000 000 L.)

(a) Rewrite reservoir P's equation in the form $y = mx + b$ and state its rate of change with units.

(b) Convert reservoir Q's rate to megalitres per week.

(c) Which reservoir is filling faster, and by how much? Give the difference in ML per week and again in litres per week.

(d) Explain why comparing 3.5 with 520 000 directly is not a valid comparison, even though it happens to pick the right reservoir, and state the one step that had to happen first.

(e) Interpret reservoir P's y-intercept in a sentence with units.

18. *(Multiple Choice)* A graph of the mass of a helium cylinder against the number of balloons filled from it passes through (40, 18.4) and (120, 14.0), where mass is in kilograms. The rate of change and the mass of the full cylinder are, respectively,
A. -0.055 kg per balloon and 16.2 kg
B. -0.055 kg per balloon and 20.6 kg
C. 0.055 kg per balloon and 20.6 kg
D. -18.2 kg per balloon and 20.6 kg

19. *(Numerical Response)* When the power fails, the air in a commercial freezer warms at a constant rate. Twenty-five minutes after the failure the temperature is $-15°$C, and fifty-five minutes after the failure it is $-6°$C. The number of minutes after the failure at which the temperature reaches 0°C is _____.
(Answer as a three-digit number.)

Lesson 7 — Extra Practice

Equations of Linear Relations Unit Review

Read this before you start. *This is the unit's review set, so nothing here tests one lesson on its own — every question below pulls together at least two of Lessons 1–6. Three conventions are being marked throughout.* ***General form*** *means* $Ax + By + C = 0$ *with* ***integer*** *coefficients and* A ***positive****. A vertical line's slope is* ***undefined*** *— never "no slope", never* ∞*.* ***Intercepts are ordered pairs****, and* ***a rate in context is a sentence carrying units****.*

Part A — One Line, Every Form

1. **The whole unit on one line.** Everything Lessons 1 through 6 taught you can be asked about the single equation

$$3x + 4y - 24 = 0.$$

Work through the parts in order; each one feeds the next.

(a) State the slope using $m = -\dfrac{A}{B}$, then confirm it by solving the equation for y.

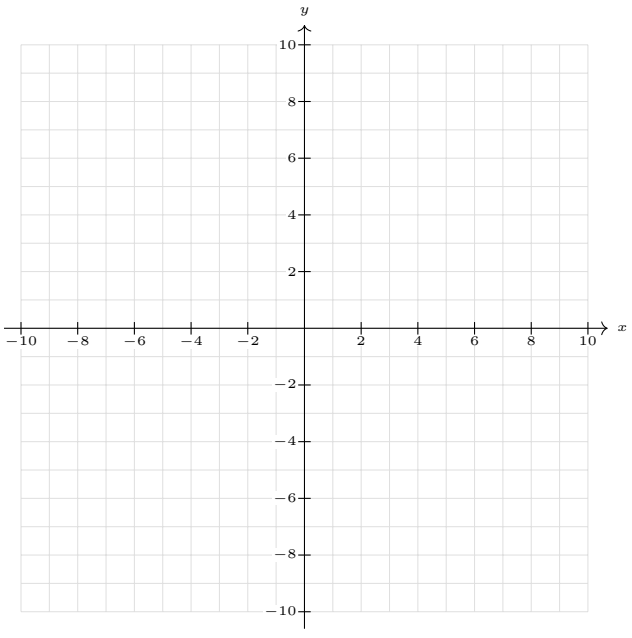
(b) Determine both intercepts **as ordered pairs**, showing the substitution for each.

(c) Write the equation in slope y-intercept form, $y = mx + b$.

(d) Decide by substitution — not by graphing — whether $P(-4, 9)$ and $Q(5, 2)$ lie on the line. For the one that does not, determine the y-value the line really has at that x, and say how far the point sits from the line vertically.

(e) Write **two** point-slope equations, $y - y_1 = m(x - x_1)$: one using the x-intercept as (x_1, y_1), and one using whichever of P and Q turned out to be on the line. Then convert both to general form and confirm they agree.

(f) Graph the line on the grid below, labelling both intercepts as ordered pairs.
(j) *(Come back for this part.)* Draw your answers to (g) and (h) on the **same** grid, and label them.



(g) Determine the equation, in general form, of the line through $(-4, -5)$ that is **parallel** to the given line.

(h) Determine the equation, in general form, of the line through $(6, -1)$ that is **perpendicular** to the given line.

(i) Confirm (h) by multiplying the two slopes. State the product and what it proves.

(k) **Now give the line a job.** On a sea-kayak day trip, x is the number of kilometres paddled and y is the number of litres of drinking water still in the group's supply.

(k1) Interpret the slope in a sentence carrying units.

(k2) Interpret each intercept in a sentence carrying units. Say what event the x-intercept marks.

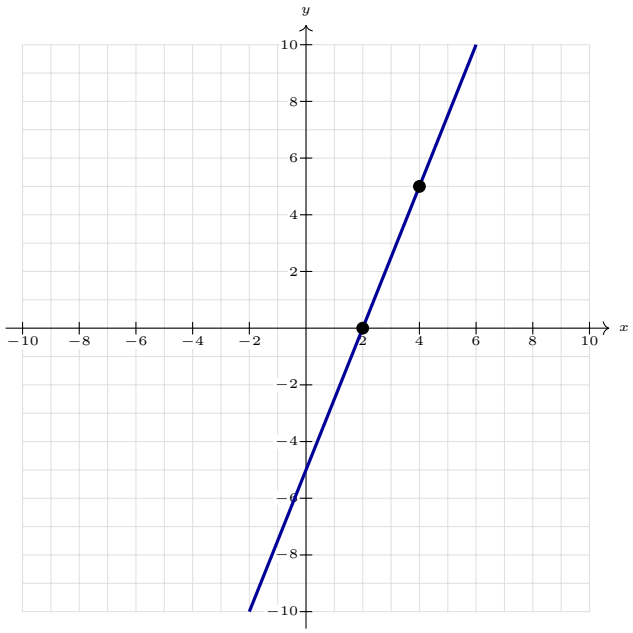
(k3) How much water is left after 5 km? After how many kilometres is the supply gone? Name the operation each of those two questions needs.

(l) Three forms of this one line have now been written. Say which form answered (a) and (c) with the least work, which answered (b) with the least work, and which answered (d) with the least work. A single sentence on each is enough.

2. **One line, four descriptions.** The four descriptions below are all supposed to be the same line. Your job is to check.

I. The equation $5x - 2y - 10 = 0$.

II. The graph



III. The table

x	-2	0	2	4	6
y	-10	-5	0	5	10

IV. The two points $(-2, -10)$ and $(6, 10)$.

(a) From **I**, determine the slope and both intercepts as ordered pairs.

(b) From **II**, read the slope and the y-intercept off the two marked points. Do they match (a)?

(c) From **III**, determine the slope. **The x-values step by 2, not by 1** — say what that does to the first differences.

(d) From **IV**, build the equation in point-slope form and convert it to general form. Does it match **I**?

(e) Only one of the four descriptions hands you the y-intercept with no work at all. Which, and why?

(f) A classmate adds a fifth description: "slope $\frac{5}{2}$, through $(0, 5)$." Write its general form and state exactly how it is related to the other four.

Part B — Building From a Condition

Not one line below hands you a slope and a point. Each description hides one or both, and some hide them inside another line. ***Write down the slope and the point you extracted before you touch a formula.***

3. Determine the equation of each line in **general form**, $Ax + By + C = 0$.

(a) through the midpoint of $A(-7, 2)$ and $B(3, -8)$, perpendicular to $4x - y + 6 = 0$

(b) through the y-intercept of $3x - 5y + 20 = 0$, parallel to the line joining $(-6, 1)$ and $(2, -3)$

(c) the **perpendicular bisector** of the segment joining $C(-5, 6)$ and $D(7, 2)$

(d) a line with non-zero intercepts whose x-intercept is exactly **three times** its y-intercept, passing through $(9, 4)$

(e) through $(-3, 5)$, parallel to the x-axis; **and** through $(-3, 5)$, perpendicular to the x-axis

(f) One of your two answers to (e) has $A = 0$ and so cannot have "A positive". State what the unit does instead in that case, and which coefficient is made positive.

(g) Verify (c) by showing that C and D are the same distance from the point $(2, 7)$ on your bisector.

4. Consider the line $kx - 6y + 18 = 0$, where k is a constant.

(a) Rearrange into $y = mx + b$, carrying k as a letter. State m and b, and say which of the two does not depend on k.

(b) Determine k so that the line is **parallel** to $2x + 3y - 9 = 0$. Then write your line in the unit's general form and look hard at the result.

(c) Part (b) has a sting in it. Say what went wrong with the word "parallel", and change the constant 18 to a value that makes the line parallel to **and different from** $2x + 3y - 9 = 0$.

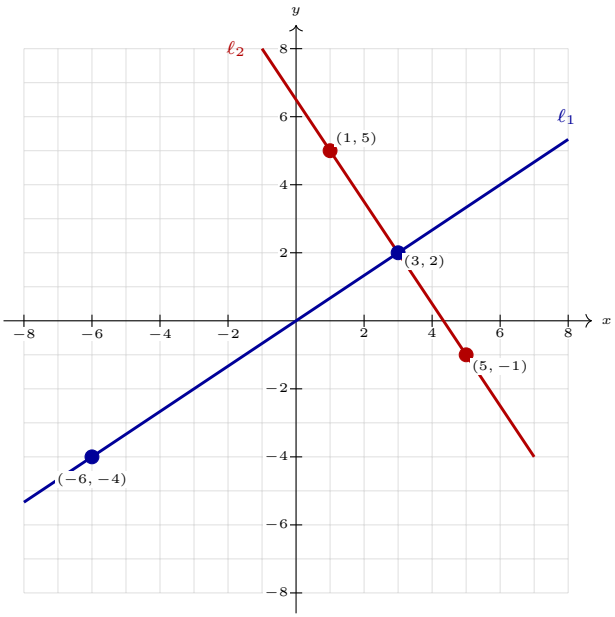
(d) Determine k so that the line is **perpendicular** to $2x + 3y - 9 = 0$, and give the answer in lowest terms in general form.

(e) Determine k so that the line passes through $(-4, 1)$, and separately determine k so that the line has x-intercept $(-9, 0)$.

(f) Which is possible — making this line horizontal, or making it vertical? Give the value of k that works, and explain why the other is out of reach whatever k is.

Part C — Graphs and Tables

5. Two lines are drawn on the grid; each passes through its two marked lattice points.



(a) Write ℓ_1 in **all three** forms: slope y-intercept, general, and point-slope using one marked point.

(b) Write ℓ_2 in general form.

(c) Show in **two** ways that $\ell_1 \perp \ell_2$: once with the product of the slopes, and once by comparing the coefficients A and B of the two general forms.

(d) Read the point where the two lines cross off the graph, then **prove** it by substituting into both equations.

(e) ℓ_1 has $C = 0$. State what that always tells you about a line, and what it does to the two intercepts.

(f) One of these two lines cannot be graphed by the two-intercept strategy. Say which and why, and name the strategy you would use for it instead.

6. The table below comes from a linear relation. Note that x steps by 5 and that $x = 0$ is not in the table.

x	−6	−1	4	9	14
y	13	9	5	1	−3

(a) A student reads the y-row, sees it dropping by 4, and writes $m = -4$. Explain the error and determine the correct slope.

(b) Determine b, showing the substitution. Why can it not simply be read off this table?

(c) Write the relation in all three forms.

(d) Determine y when $x = 34$.

(e) Determine x when $y = 25$.

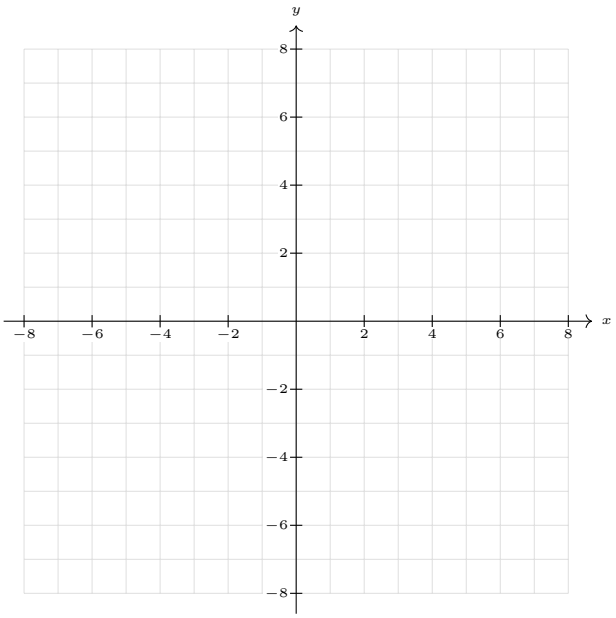
(f) Say which form you used in (d) and which in (e), and why they were not the same form.

7. **A family of three lines on one grid.** Let ℓ be the line $5x + 2y + 10 = 0$.

(a) State both intercepts of ℓ as ordered pairs and graph ℓ on the grid below.

(b) Determine, in general form, the line p through $(2, 3)$ **parallel** to ℓ, and graph it.

(c) Determine, in general form, the line q through $(2, 3)$ **perpendicular** to ℓ, and graph it.



(d) State the one point p and q have in common, without doing any algebra, and say how you know.

(e) Your three general forms share a pattern. State the rule for writing a parallel line and the rule for writing a perpendicular line **straight from the coefficients** A and B.

(f) For which of ℓ, p, q is the two-intercept strategy the clean one, and for which would you rather step the slope from a known point? Justify from the numbers.

8. For each equation, name the hand-graphing strategy you would use — *slope y-intercept*, *two intercepts*, or *table of values* — give the one piece of evidence that decided it, and state one point you would plot. **Do not graph anything.**

(a) $4x + 7y - 28 = 0$ (b) $y + 3 = -\frac{1}{2}(x - 6)$ (c) $2x - 7y + 5 = 0$ (d) $6x + 15 = 0$

(e) Two of the four are in general form with both an x-term and a y-term, and they do not take the same strategy. Say which two, and state the one test on A, B and C that separates them.

Part D — Slope in Context

9. A glacier's terminus is surveyed from a fixed rock marker on the valley floor. In 1995 the terminus stood 240 m from the marker; in 2015 it stood 1 060 m from it. Let t be the number of years since 1990 and D the distance from the marker in metres.

(a) Write the two surveys as ordered pairs (t, D), then determine the rate of change and interpret it in a sentence carrying units.

(b) Use point-slope form to build the model, then write it as $D = mt + b$ and interpret b in a sentence carrying units.

(c) Write the model in general form, using t for x and D for y. Check that A is positive and all three coefficients are integers.

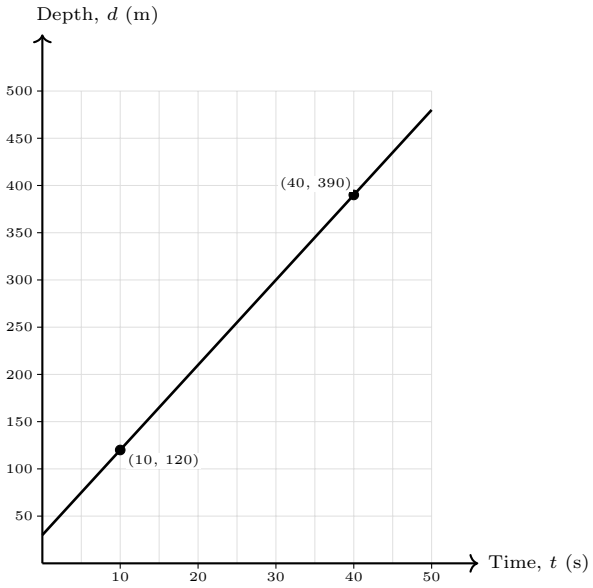
(d) Predict the distance in 2030.

(e) In which year does the model predict the terminus passes 2 km from the marker? Answer to the nearest year.

(f) Restate the rate in **kilometres per century**. Is the glacier retreating any faster in this new wording? Explain.

(g) Parts (d) and (e) run the same model in opposite directions. Name the operation each needs, and say which of your three forms you would reach for in each case.

10. The graph shows the depth d below the surface, in metres, of a hoist cage descending a mine shaft, against the time t in seconds since the descent began. **Read the tick numbers carefully — the two axes are not on the same scale.**



(a) Determine the rate of change from the two marked points and interpret it in a sentence carrying units.

(b) Write the model as $d = mt + b$, then in general form. Interpret the d-intercept in a sentence carrying units.

(c) How deep is the cage after 75 s? At what time does it reach 570 m?

(d) A second shaft's cage is modelled by $15t - 2d + 60 = 0$. State its rate with units and its starting depth.

(e) Which cage descends faster, and by how much? Which reaches 600 m first, and by how many seconds?

(f) A student compares the numbers 9 and 15 straight off the two equations and concludes the second cage is faster. Explain the mistake in one sentence, naming the step that had to happen first.

Part E — Reasoning and Error Analysis

11. Decide whether each claim is **true** or **false**. A "false" needs a counterexample or a corrected statement; a "true" needs a reason.

(a) "Parallel lines have the same equation."

(b) "A line with an undefined slope has no equation."

(c) "In $Ax + By + C = 0$, A is the x-intercept and B is the y-intercept."

(d) "Every line can be written in the form $y = mx + b$."

(e) "If the slopes of two lines are reciprocals, the lines are perpendicular."

(f) "$3x - y + 5 = 0$ and $-6x + 2y - 10 = 0$ are parallel lines."

(g) "A line with a positive slope must have a positive y-intercept."

(h) "Two lines with the same slope and the same y-intercept are the same line."

12. A student was asked for the slope and both intercepts of $6x - 9y + 12 = 0$, and then for the line through $(2, -1)$ perpendicular to it. Their whole solution reads:

$$m = -\tfrac{A}{B} = -\tfrac{6}{9} = -\tfrac{2}{3}; \qquad x\text{-int } 6, \;\; y\text{-int } -9; \qquad m_\perp = \tfrac{3}{2}; \qquad y + 1 = \tfrac{3}{2}(x - 2) \;\Rightarrow\; -3x + 2y + 8 = 0.$$

(a) Determine the true slope, and name in one sentence exactly what the student did to B.

(b) Determine both intercepts as ordered pairs, and say which coefficient the student ignored completely.

(c) Determine the correct perpendicular line in general form.

(d) The student's final equation passes through $(2, -1)$ — check it. Explain why that check therefore catches neither of their two real errors, and state a check that catches both.

(e) Even if every slope had been right, the student's last line would still have been marked down. State why, and repair it.

13. Asked for the equation of the line through $(-3, 5)$ perpendicular to $x = 7$, a student writes: "$x = 7$ is a line with slope 0, so the perpendicular has an undefined slope, and the answer is $x = -3$."

(a) State the slope of $x = 7$ using the correct word, and the slope of the line the question actually wants.

(b) Give the correct equation, in general form, and state how the house rule is applied to it.

(c) The student's line $x = -3$ is a real line. State precisely how it is related to $x = 7$, and how far from it.

(d) Explain why the test $m_1 m_2 = -1$ cannot be applied to this pair at all, and state the wording the unit uses for this exception.

Part F — Challenge

14. **A triangle delivered as three equations.** The three lines

$$\textbf{L1: } 2x + y - 10 = 0 \qquad \textbf{L2: } x - 2y - 5 = 0 \qquad \textbf{L3: } x + 3y - 15 = 0$$

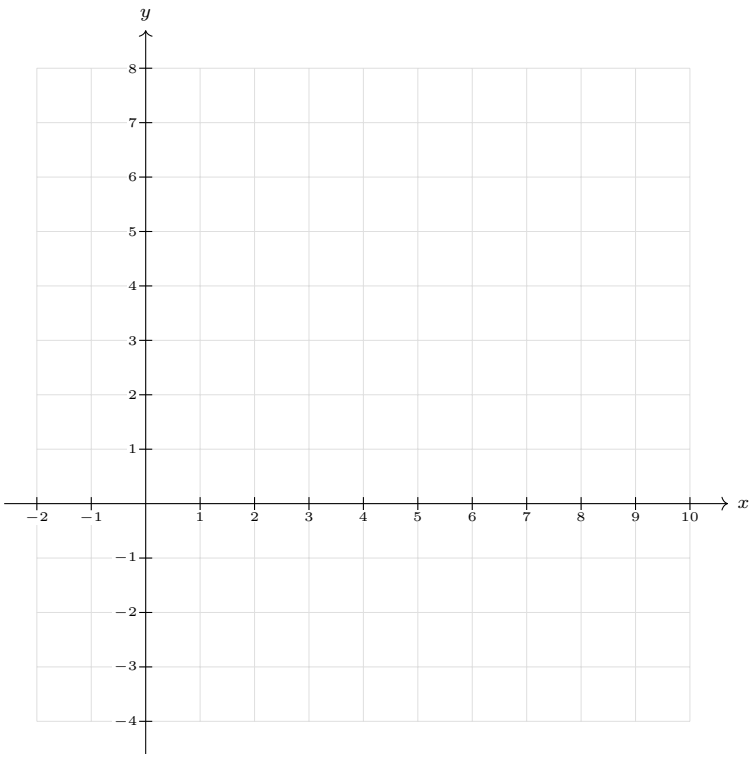
bound a triangle.

(a) Determine the three vertices **algebraically**, naming the pair of lines that gives each one.

(b) State the slope of each line, and identify the pair that meets at 90°. Which vertex is the right angle at?

(c) Use the distance formula on all three sides. What kind of triangle is this? Verify with the Pythagorean relationship.

(d) Graph all three lines on the grid and label them L1, L2, L3. Your three vertices from (a) must land on the intersections you draw.

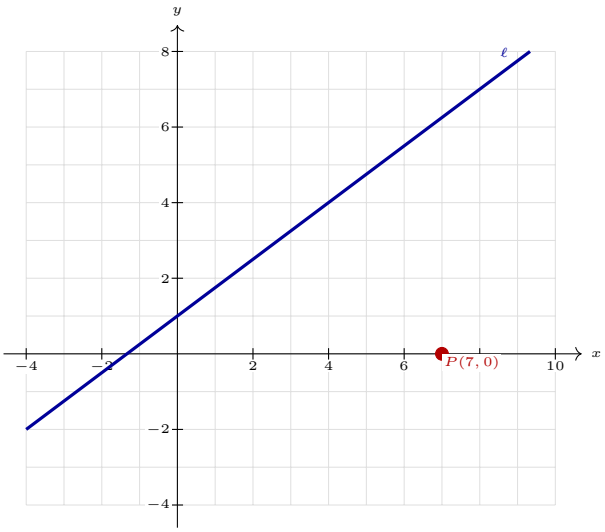


(e) Determine the area of the triangle, in square units, using the two legs of the right angle as base and height.

(f) Determine, in general form, the line from the right-angle vertex to the midpoint of the opposite side.

(g) Show that the line in (f) is also perpendicular to that side. What does that tell you about the triangle, and which two names for the same segment have now both been earned?

15. **How far is a point from a line?** "Distance from a point to a line" always means the *perpendicular* distance — the shortest of all the distances from the point to points of the line. Let ℓ be $3x - 4y + 4 = 0$ and let $P(7, 0)$.



(a) Determine the equation, in general form, of the line through P perpendicular to ℓ.

(b) Determine the point F where that perpendicular meets ℓ, by solving the two equations together.

(c) Use the distance formula to find PF. This is the distance from P to ℓ.

(d) The point $G(8, 7)$ is also on ℓ. Compute PG and confirm it is longer than PF, as it must be.

(e) Check your answer with the formula $\dfrac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$, which is this whole calculation packaged up.

(f) Determine the point P' obtained by continuing from P through F the same distance again — the mirror image of P in ℓ. Verify that P' is on your perpendicular and the same distance from ℓ as P is.

16. *(Multiple Choice)* The equation, in general form, of the line through $(-2, 9)$ parallel to $7x + 3y - 15 = 0$ is
A. $7x + 3y + 13 = 0$ B. $7x + 3y - 13 = 0$ C. $3x - 7y + 69 = 0$ D. $7x + 3y - 41 = 0$

17. *(Multiple Choice)* Four lines are given.

$$i)\ 6x - 4y + 7 = 0 \qquad ii)\ 2x + 3y - 12 = 0 \qquad iii)\ 9x - 6y - 5 = 0 \qquad iv)\ 4x + 6y + 9 = 0$$

Which statement is true?
A. $i) \parallel iii)$ only B. $ii) \parallel iv)$ only
C. $i) \parallel iii)$, $ii) \parallel iv)$, and $i) \perp ii)$
D. $i) \parallel iii)$ and $ii) \parallel iv)$, but no pair is perpendicular

18. *(Numerical Response)* The line through $(-5, 2)$ perpendicular to $3x+7y-21=0$ is written in general form $Ax+By+C=0$, where A, B and C are integers with A positive and no common factor greater than 1. The value of $A+B+C$ is _____. *(Answer as a three-digit number.)*

Part G — Two Skills This Review Had Not Yet Touched

19. **Is it even a line?** Lesson 1's first question was never "what is the slope" — it was "is this a linear relation at all". Sort every letter below into exactly one of the three rows, rearranging first where you need to. Then, for every letter in rows 1 and 3, write the relation in **general form**.

(a) $3y = 12 - 2x$ (b) $y = \frac{5}{x} - 1$ (c) $4x + 9 = 0$ (d) $y = \frac{2x-7}{4}$

(e) $y^2 = x + 3$ (f) $y + 6 = 0$ (g) $\frac{x}{5} - \frac{y}{2} = 1$ (h) $y = x(x-4)$

LINEAR (can be written $y = mx + b$): _____

NON-LINEAR (the graph is not a straight line): _____

STRAIGHT LINE, but **not** of the form $y = mx + b$: _____

(i) Exactly one of your general forms has $B = 0$ and exactly one has $A = 0$. Name them, and say what each missing coefficient does to the graph.

(j) State the domain and the range of the relations in (c) and (f).

(k) Parts (c) and (f) are perpendicular to each other. Explain why you cannot prove that with $m_1 m_2 = -1$.

20. **Average rate against constant rate.** Pond A and pond B are both monitored for algae. Pond A's readings are in the table; pond B grows at a **constant** rate, starting at 4 kg at $t = 0$ and reaching 64 kg at $t = 8$ weeks.

t (weeks)	0	2	4	6	8
Pond A, M (kg)	4	8	16	32	64

(a) Calculate pond A's rate of change across each of the four intervals. Is the relation linear? Justify from the numbers.

(b) Calculate pond A's **average** rate of change over the whole eight weeks and interpret it in a sentence carrying units.

(c) Write pond B's model as $M = mt + b$ and then in general form. State its rate with units.

(d) The two ponds agree at $t = 0$ and at $t = 8$. Which holds more algae at $t = 4$, and by how much?

(e) Did pond A ever actually grow at the rate you found in (b)? Answer from your four interval rates.

(f) Express pond B's rate in kilograms per day, to the nearest hundredth. Is pond B growing any faster in that wording?